KỸ THUẬT SỐ

Đánh giá

- **Điểm quá trình: 50%**
  - Chuyên cần
  - Kiểm tra
  - Tiểu luận
- **Điểm cuối kỳ: 50%**

Tài liệu tham khảo

- Nguyễn Trường Duy, Giáo trình Kỹ thuật số, Đại học Quốc gia Tp. HCM, Năm 2019.
- Ronald J. Tocci - Neal S. Widmer – Gregory L. Moss, Digital Systems: Principles and Applications, 12th Ed. Prentice Hall, 2017.
- Thomas L. Floyd, Digital Fundamental, 11th Ed. Prentice Hall, 2015.

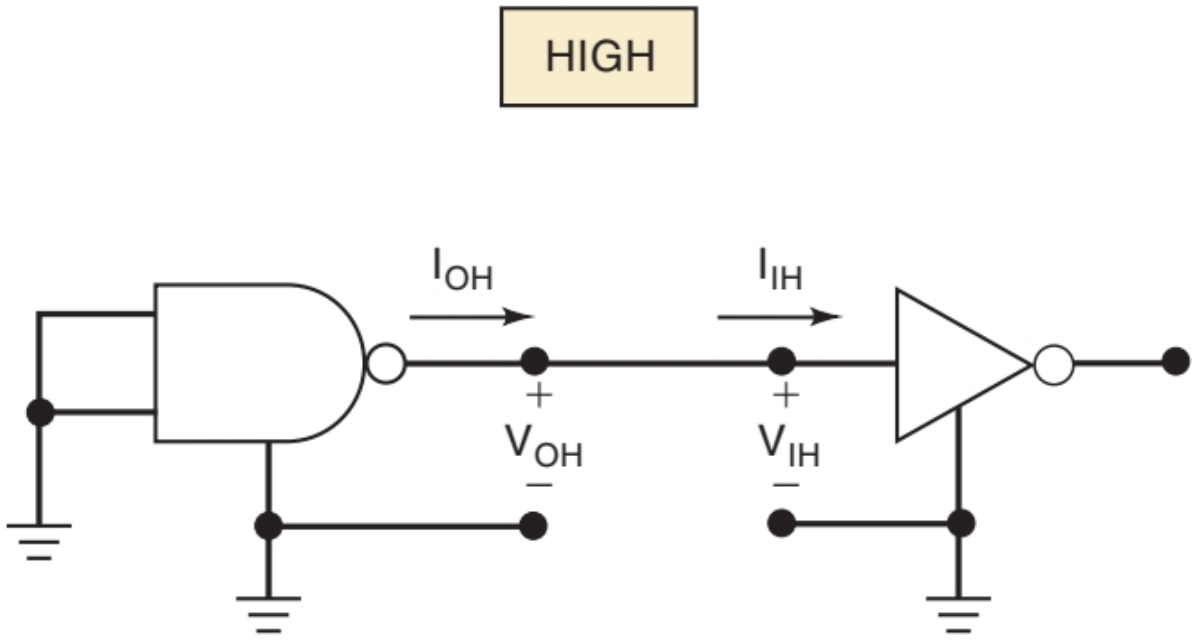
IC số

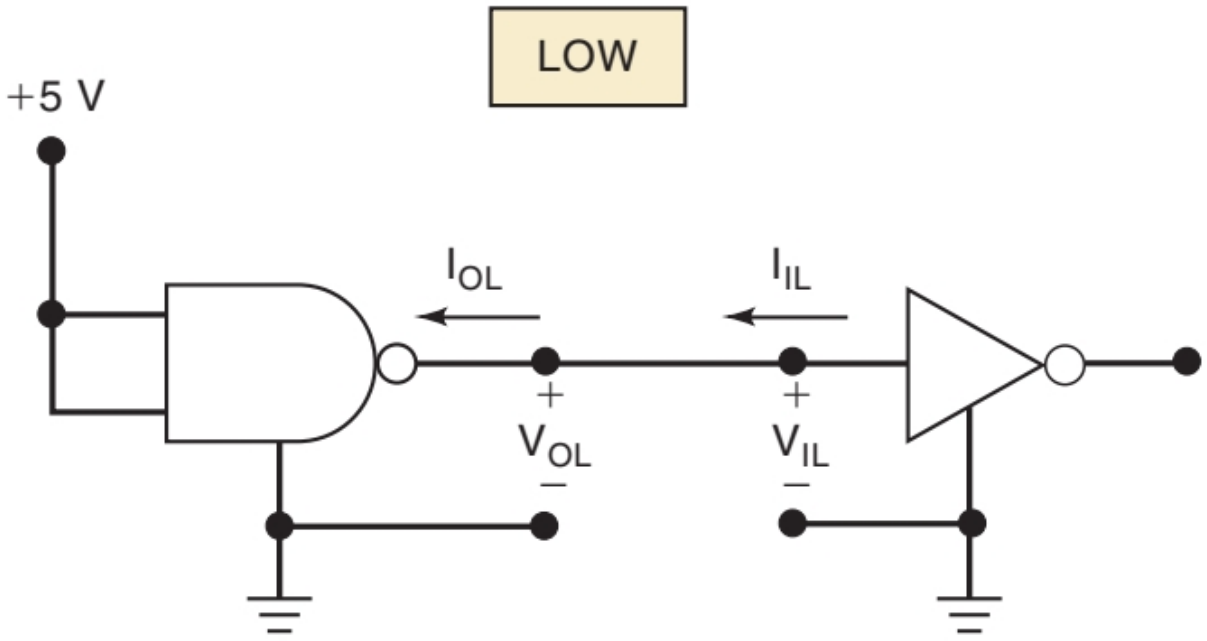
Nội dung

- Các thông số
- TTL
- CMOS
- Giao tiếp IC

Các thông số

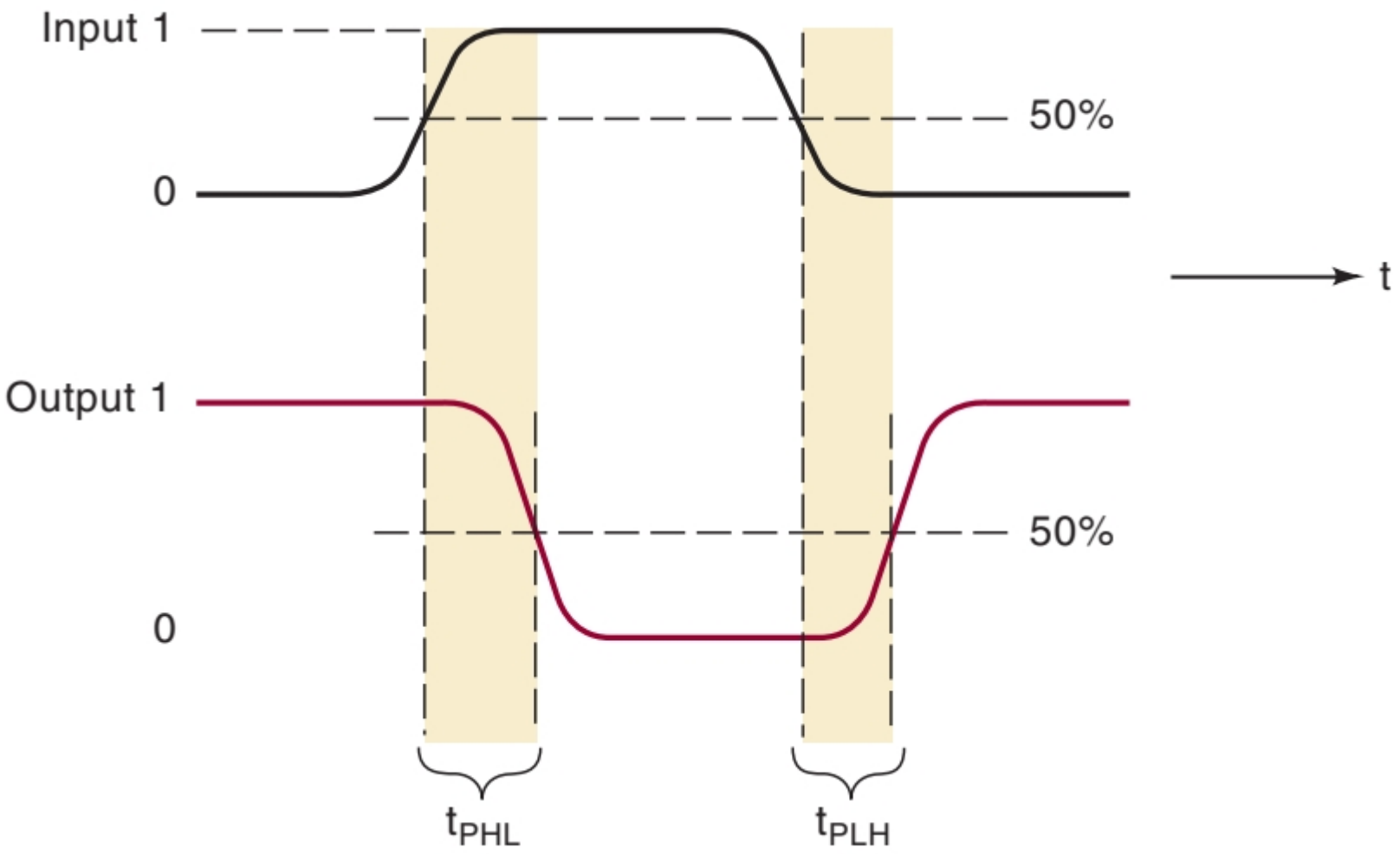
- Điện áp và dòng điện





Các thông số

- Thời gian Delay

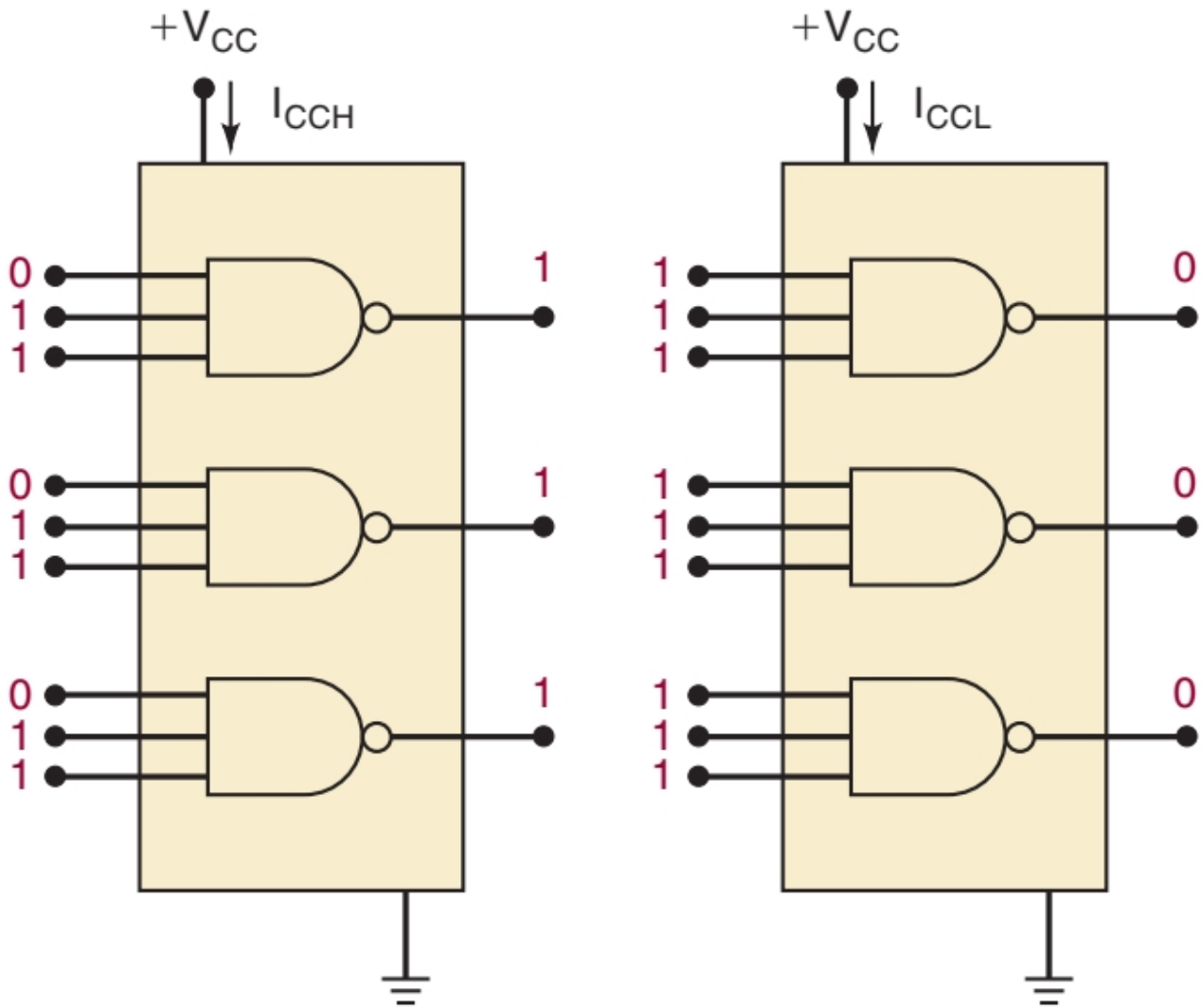


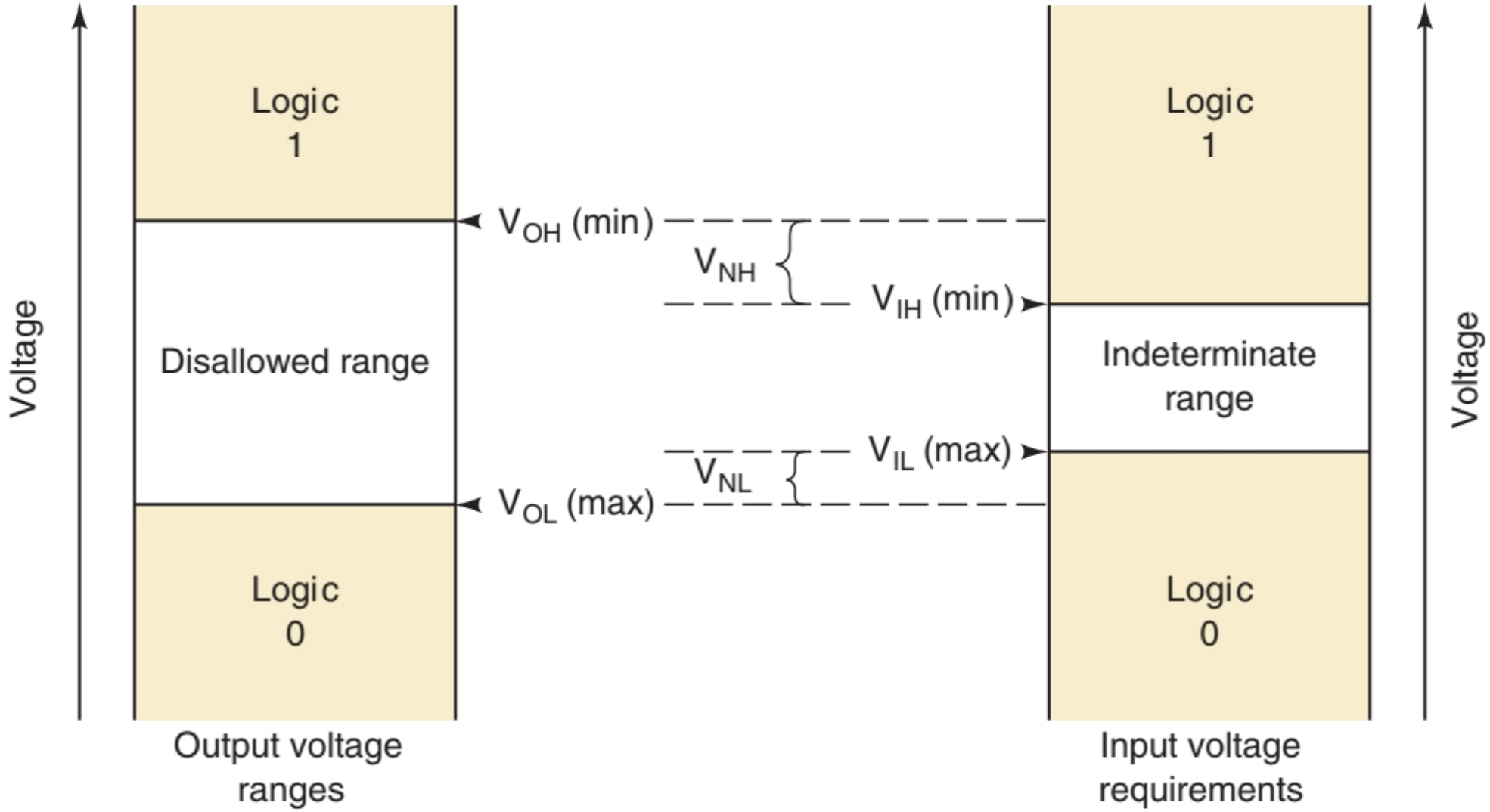
Các thông số

- Công suất tiêu tán

$$I_{CC}(\text{avg}) = \frac{I_{CCH} + I_{CCL}}{2}$$

$$P_D(\text{avg}) = I_{CC}(\text{avg}) \times V_{CC}$$



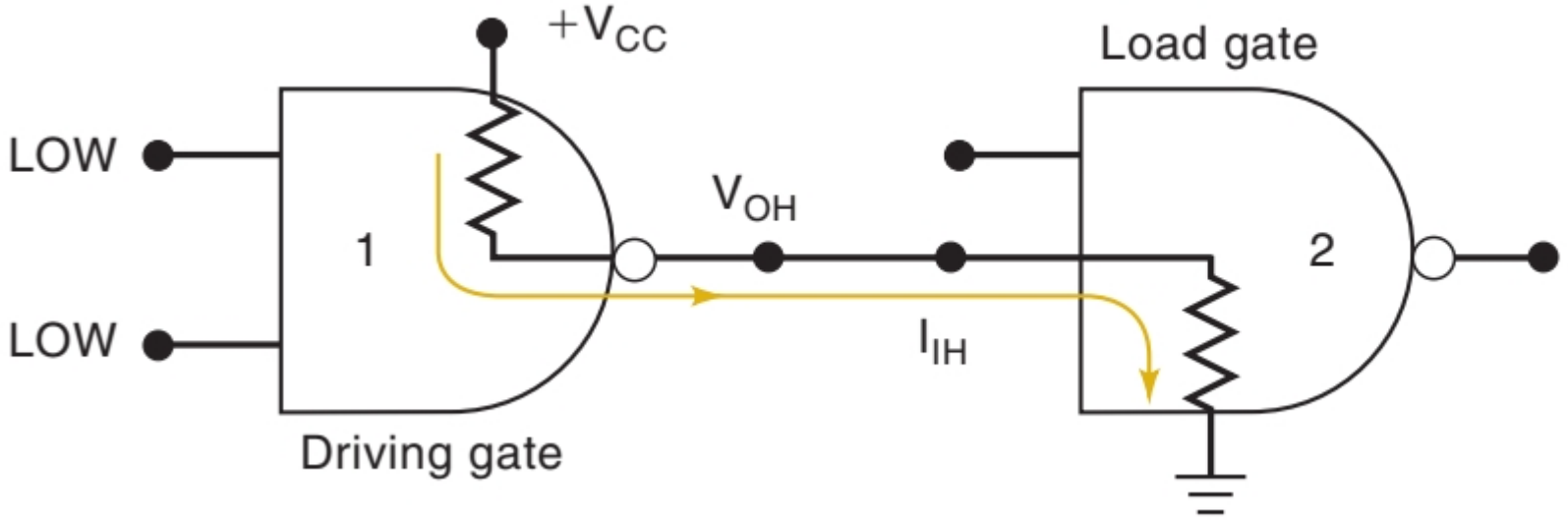


$$V_{NH} = V_{OH}(\text{min}) - V_{IH}(\text{min})$$

$$V_{NL} = V_{IL}(\text{max}) - V_{OL}(\text{max})$$

Các thông số

- Dòng nguồn

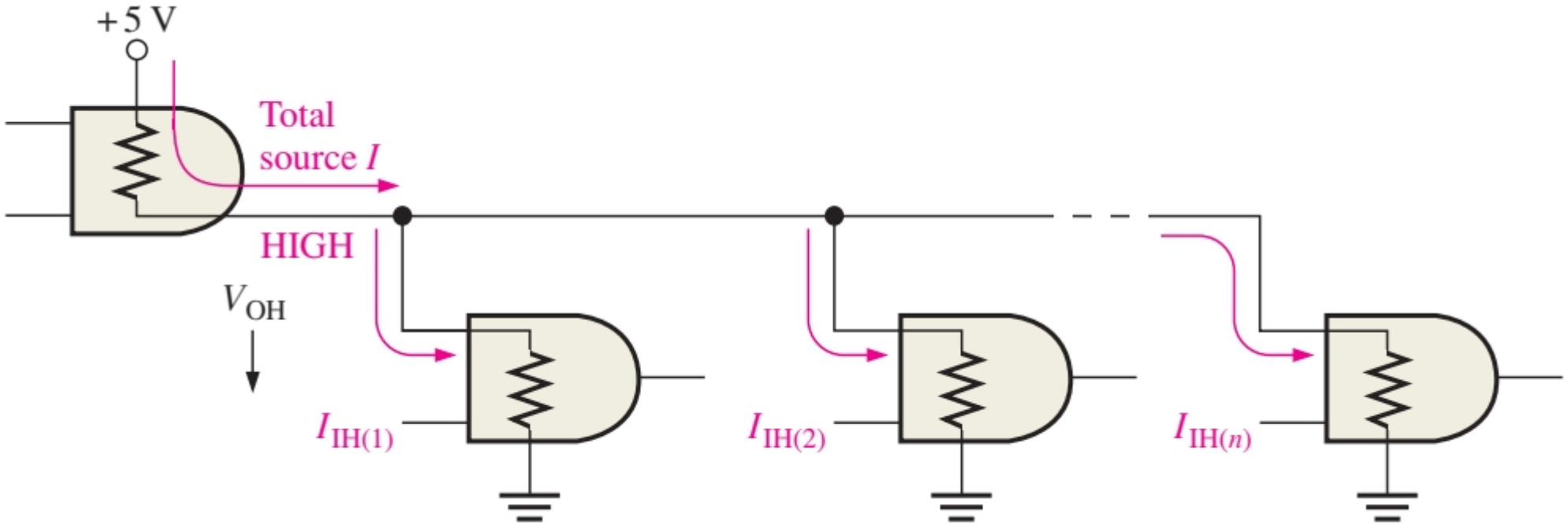


Current sourcing

Driving gate supplies (sources) current to load gate in HIGH state.

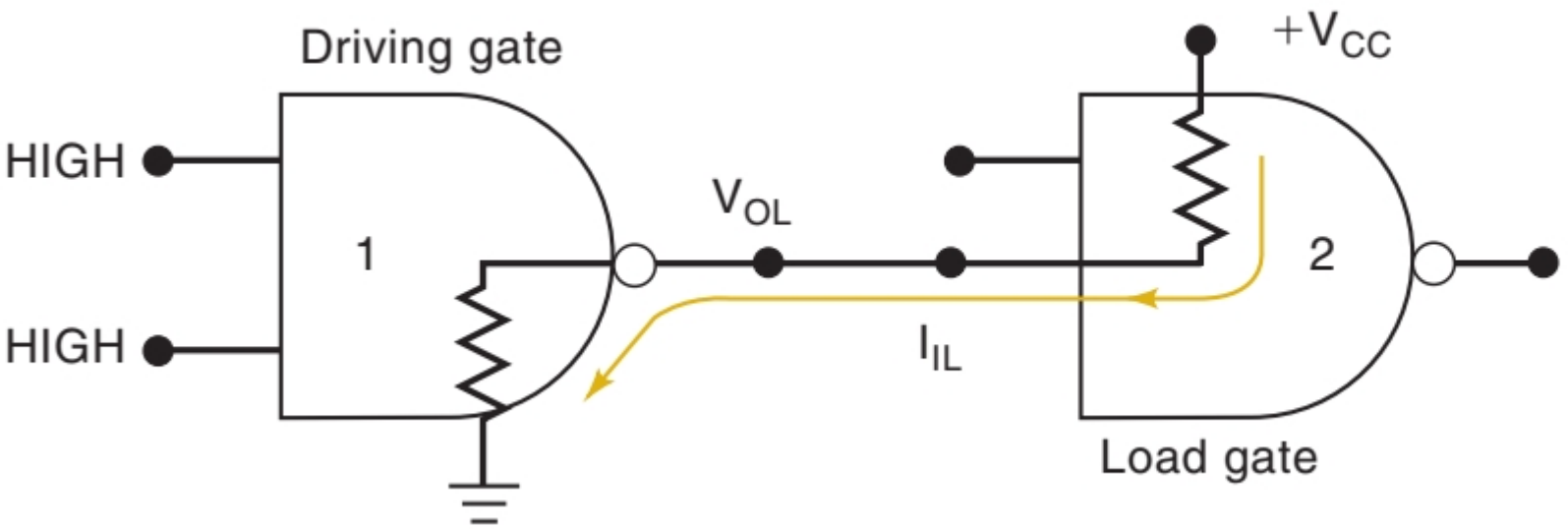
Các thông số

- Dòng nguồn



Các thông số

- Dòng đích

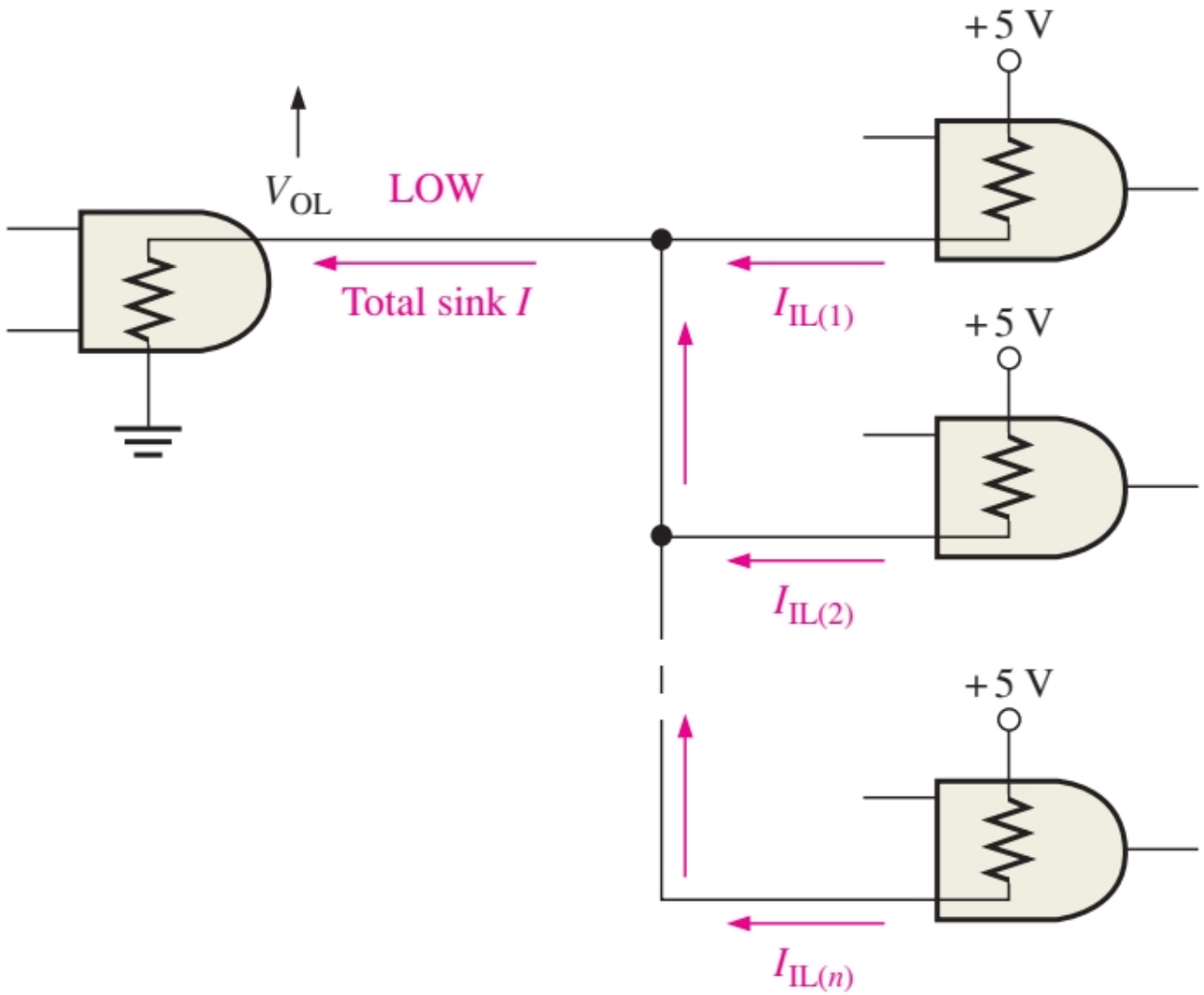


Current sinking

Driving gate receives (sinks) current from load gate in LOW state.

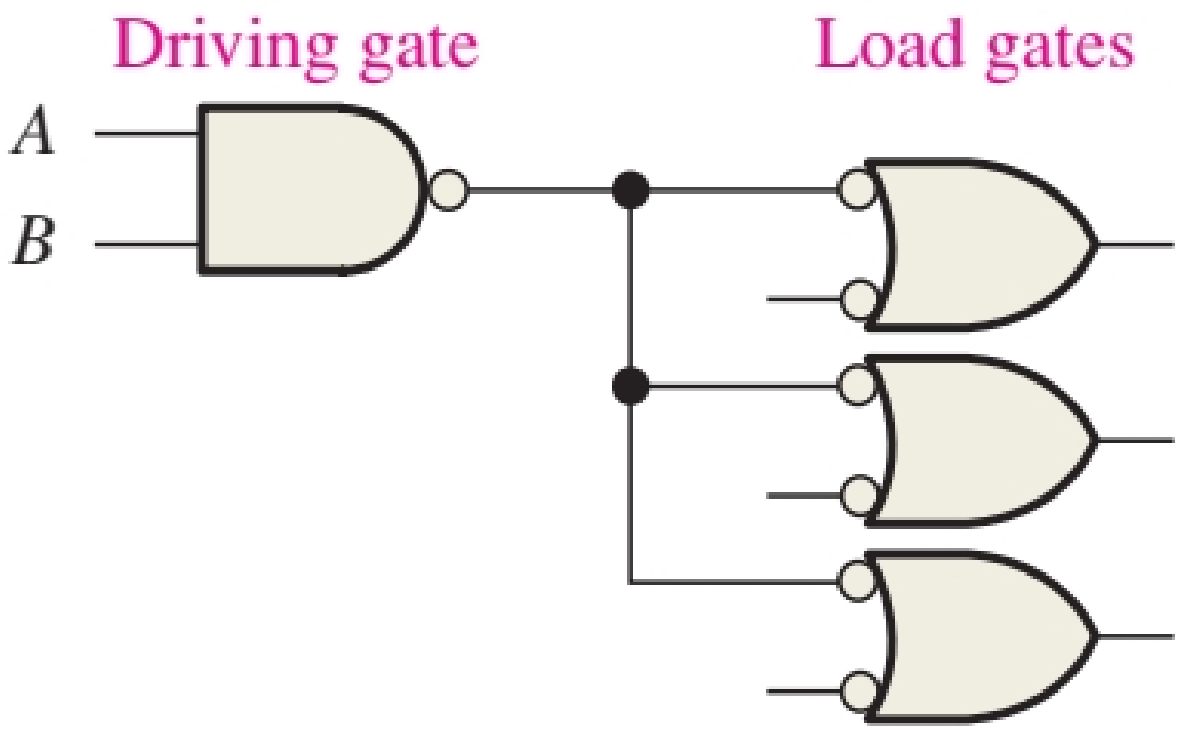
Các thông số

- Dòng đích



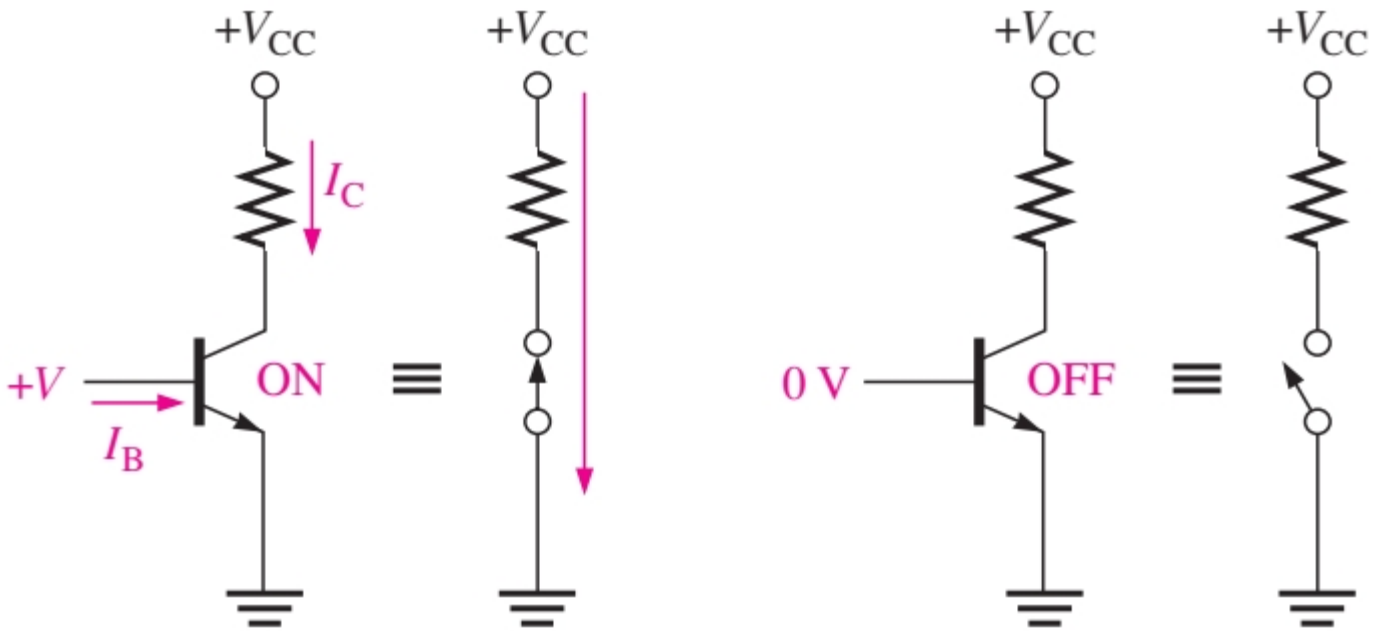
Các thông số

- Hệ số tải



TTL

- BJT

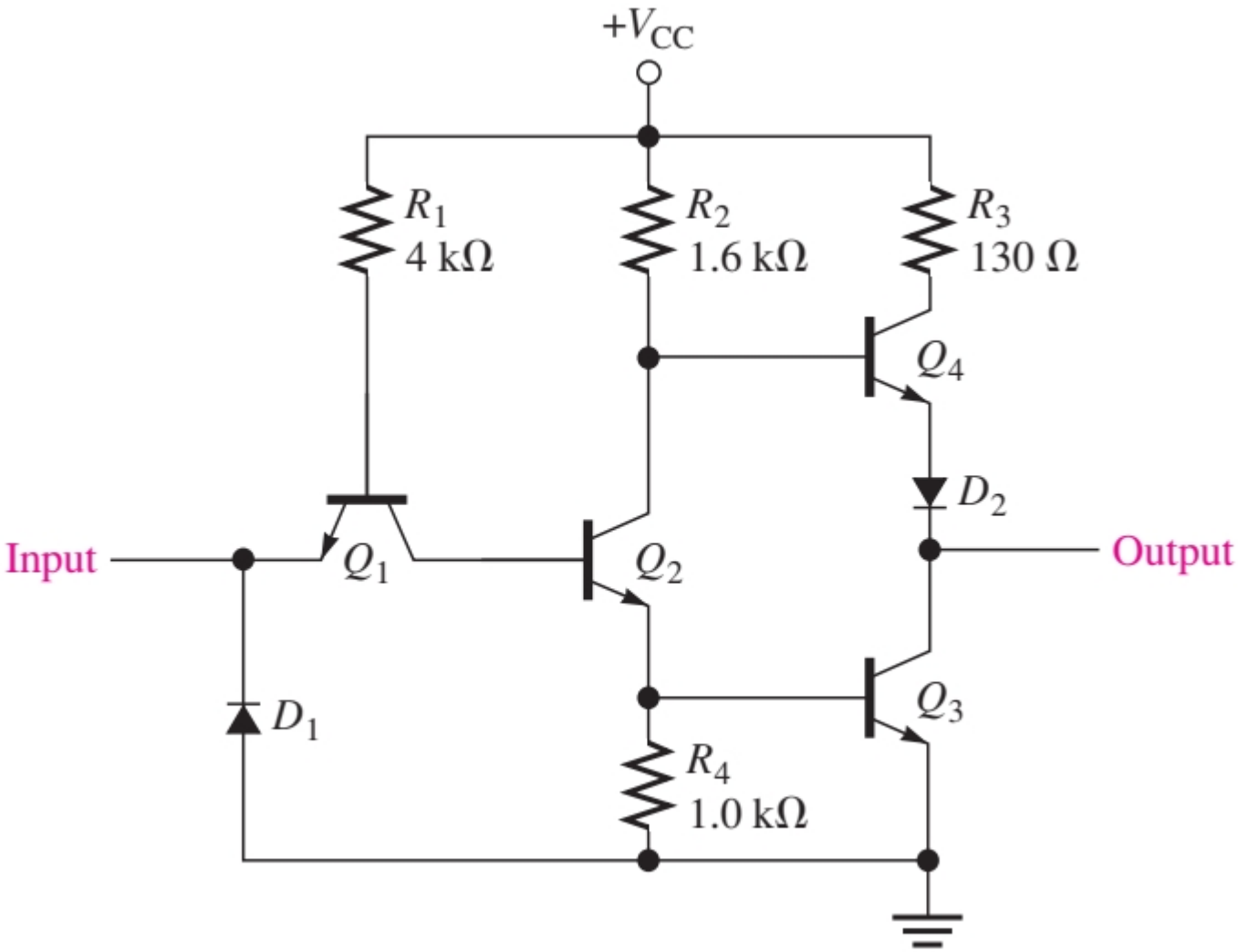


(a) Saturated (ON) transistor and ideal switch equivalent

(b) OFF transistor and ideal switch equivalent

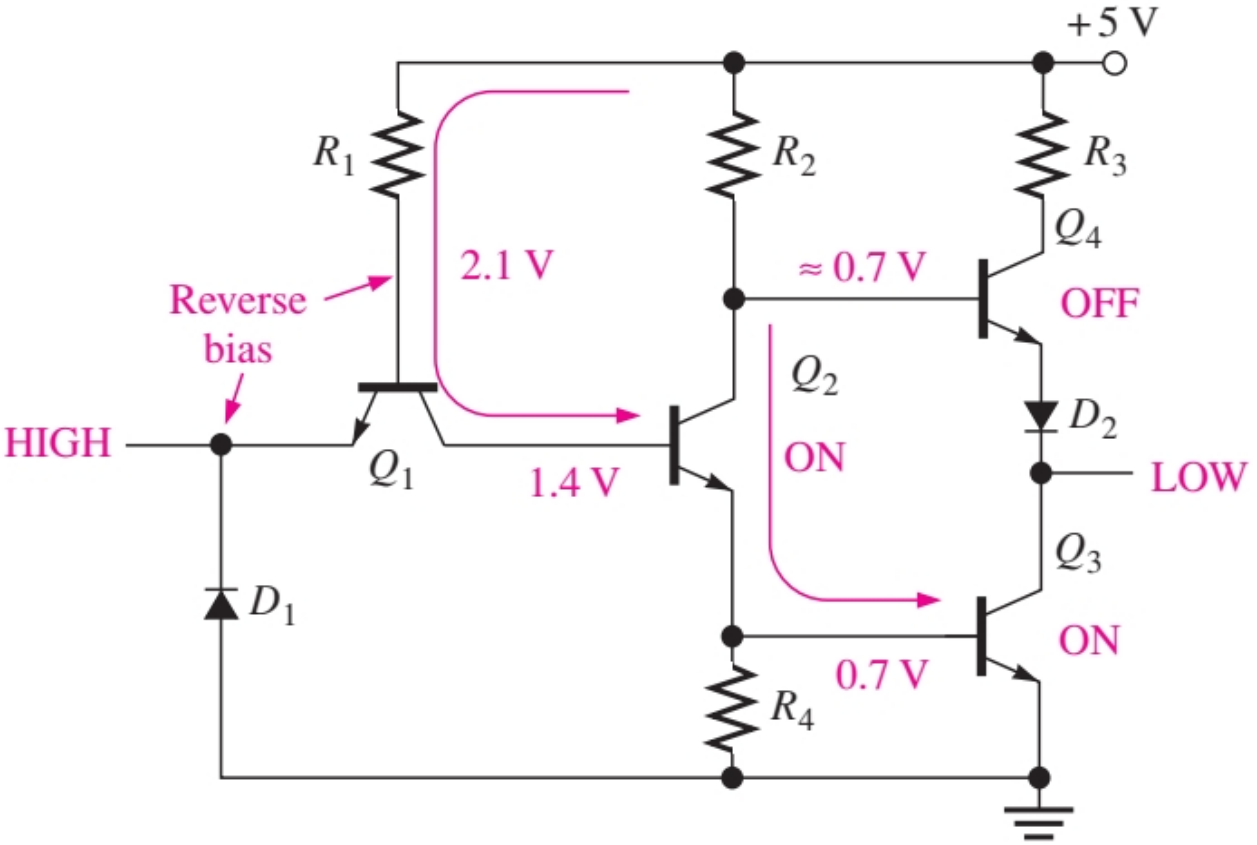
TTL

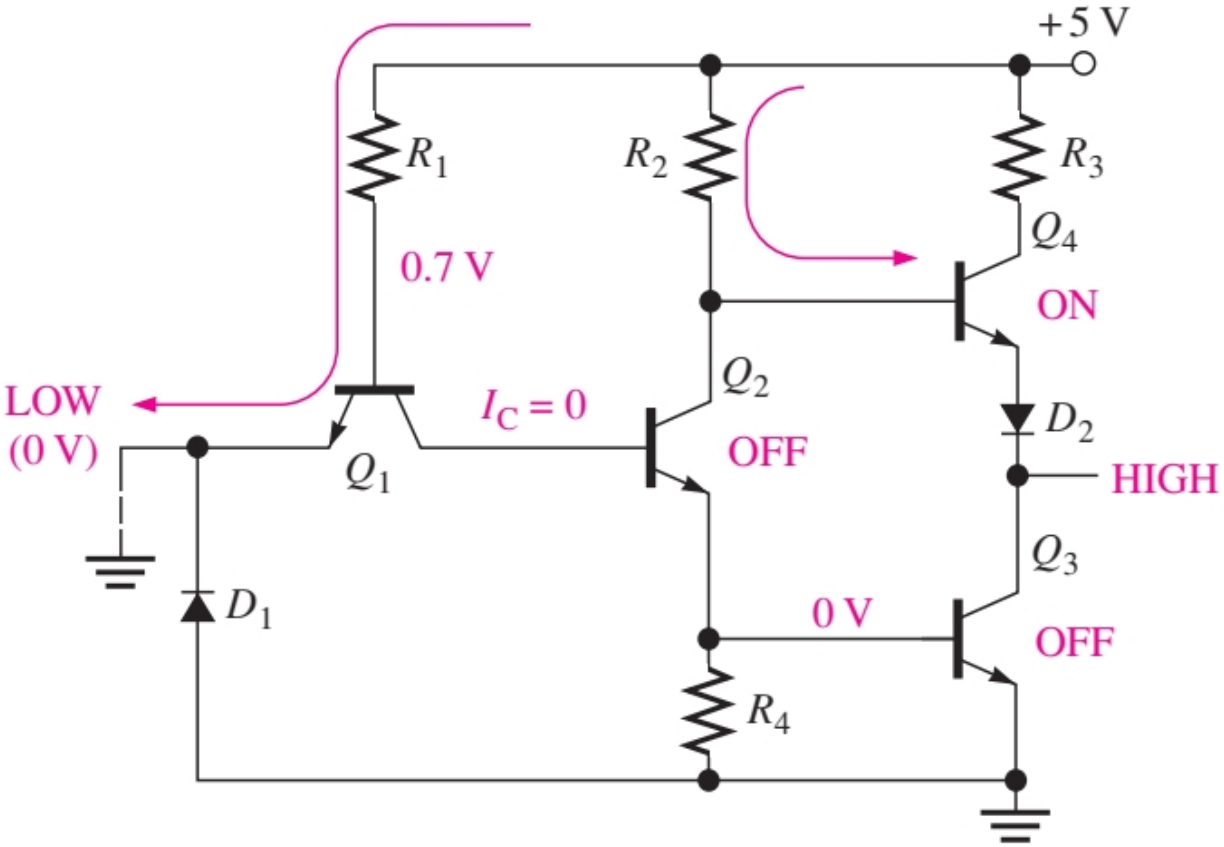
- TTL cổng NOT



TTL

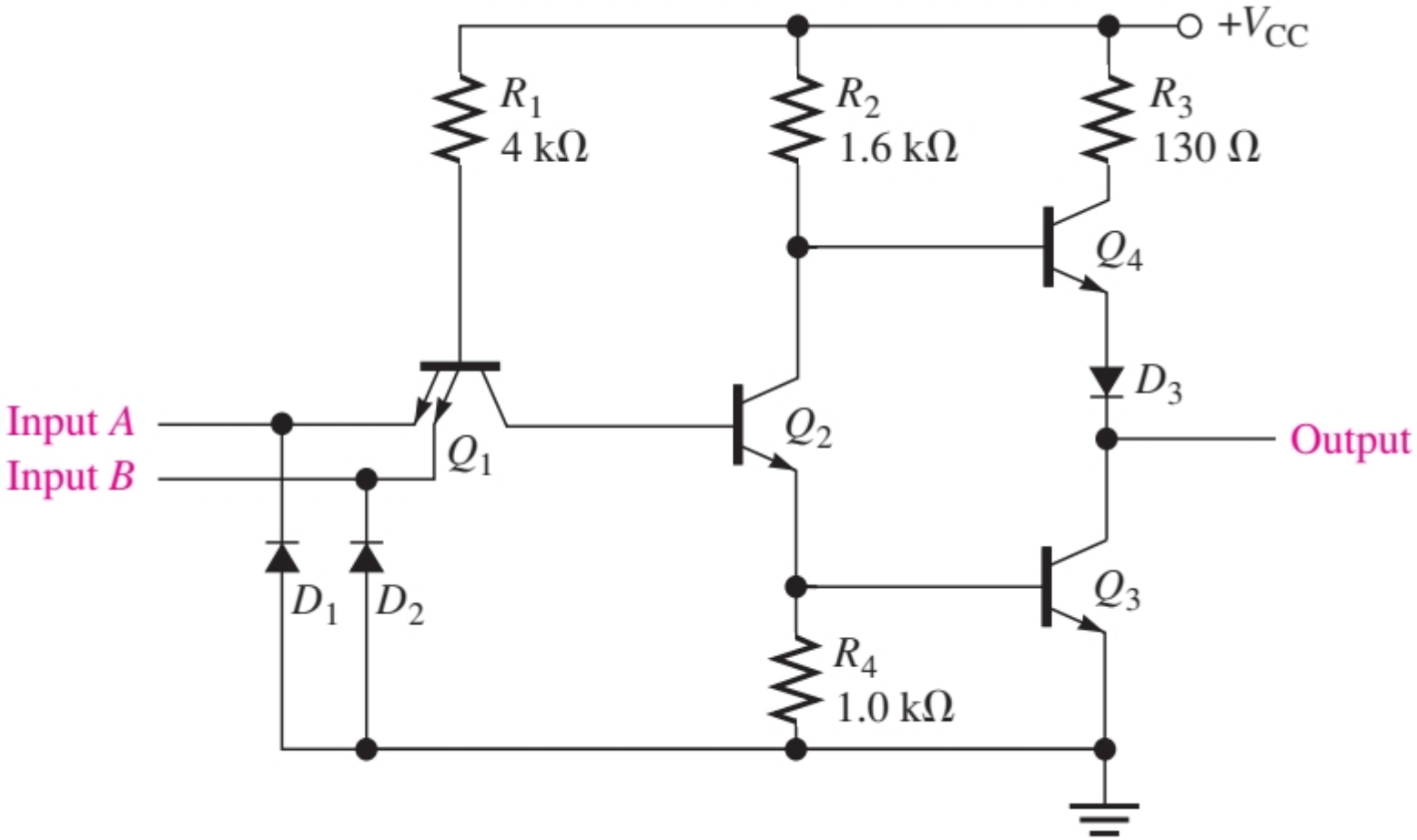
- TTL cổng NOT





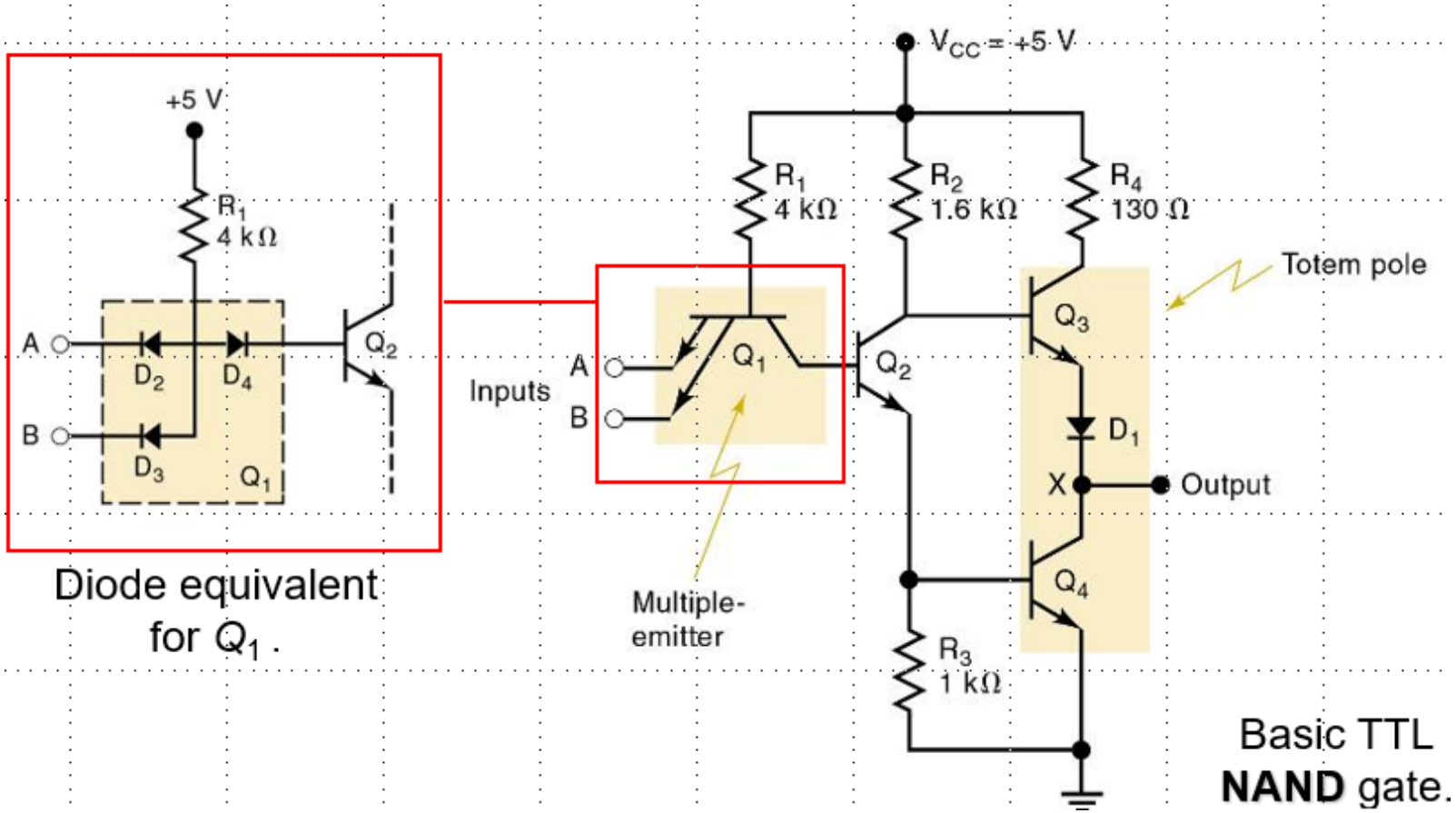
TTL

- TTL cổng NAND



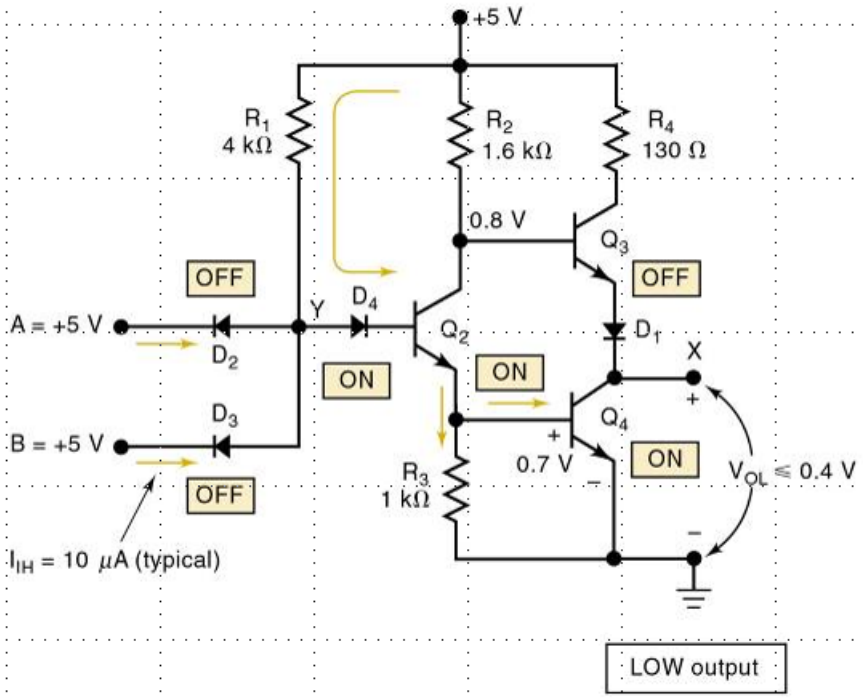
TTL

- TTL cổng NAND



TTL

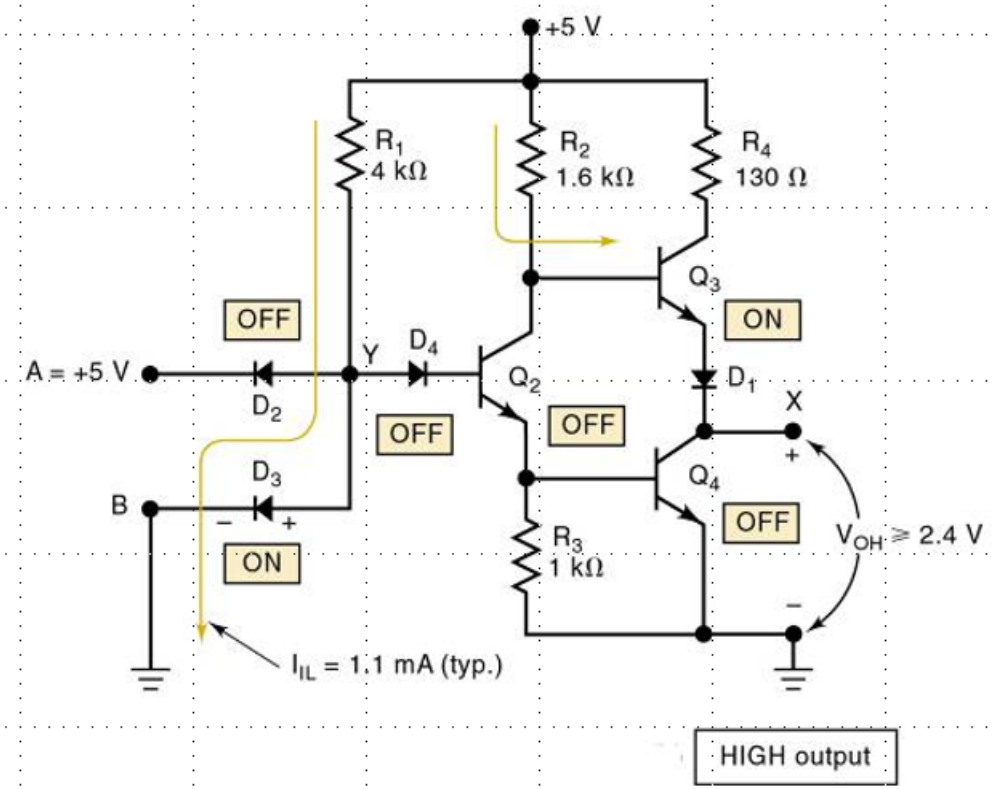
- TTL cổng NAND



Input conditions	Output conditions
A and B are both HIGH (≥ 2 V)	Q_3 OFF
Input currents are very low I_{IH} = 10 μA	Q_4 ON so that V_X is LOW (≤ 0.4 V)

TTL

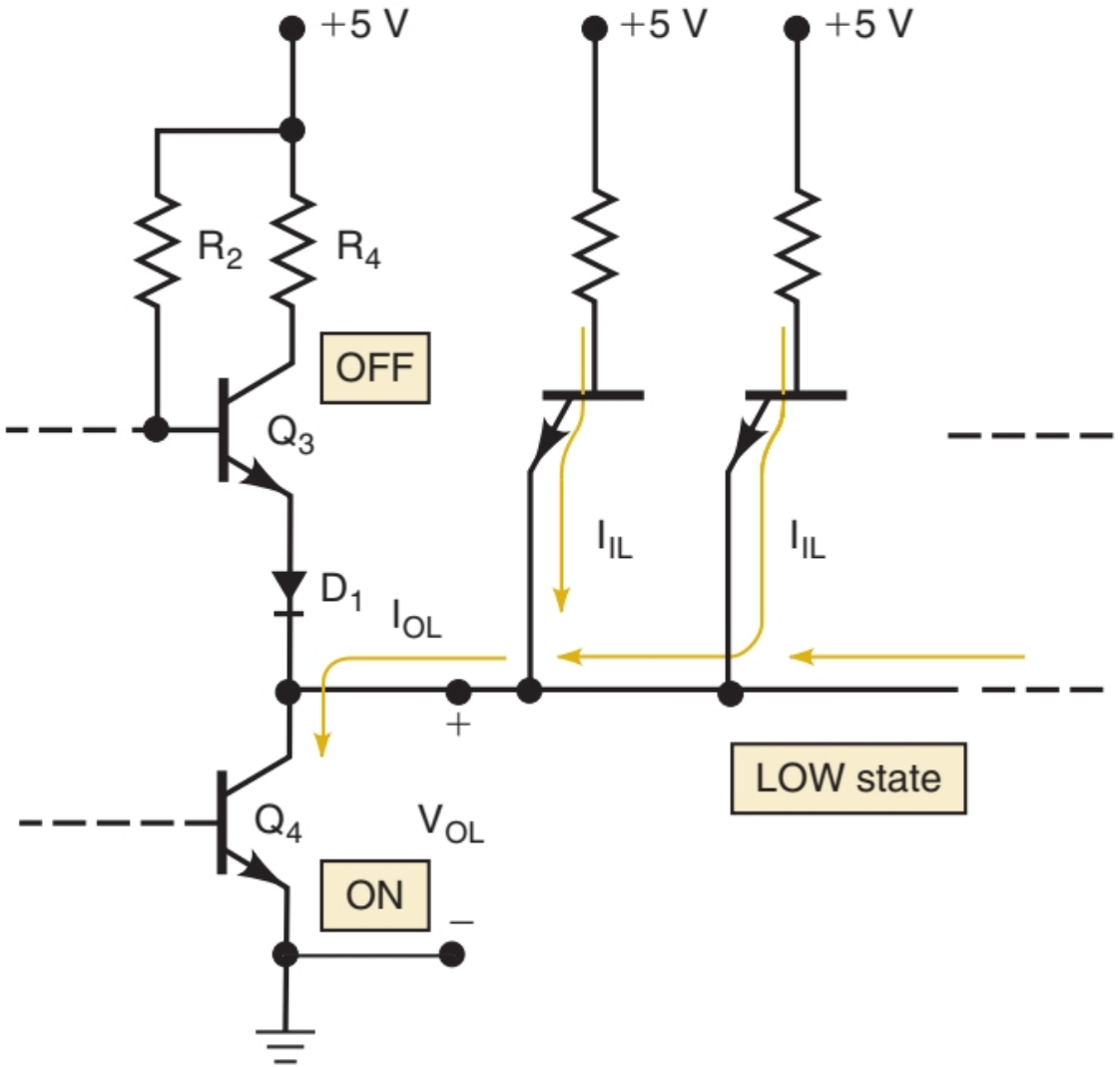
- TTL cổng NAND

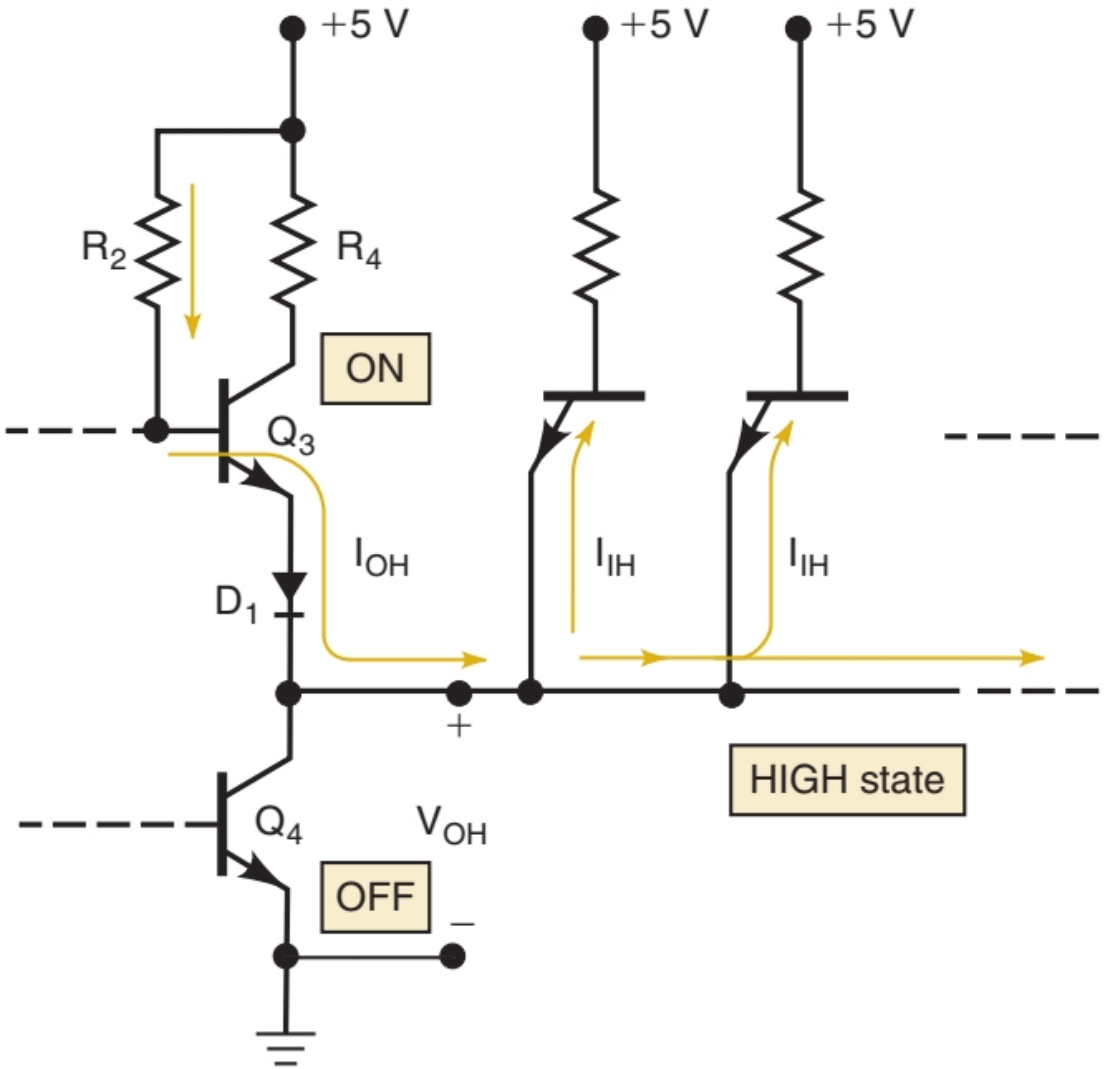


Input conditions	Output conditions
A or B or both are LOW (≤ 0.8 V)	Q_4 OFF
Current flows back to ground through LOW input terminal. I_{IL} = 1.1 mA	Q_3 acts as emitter-follower and V_{OH} ≥ 2.4 V, typically 3.6 V

TTL

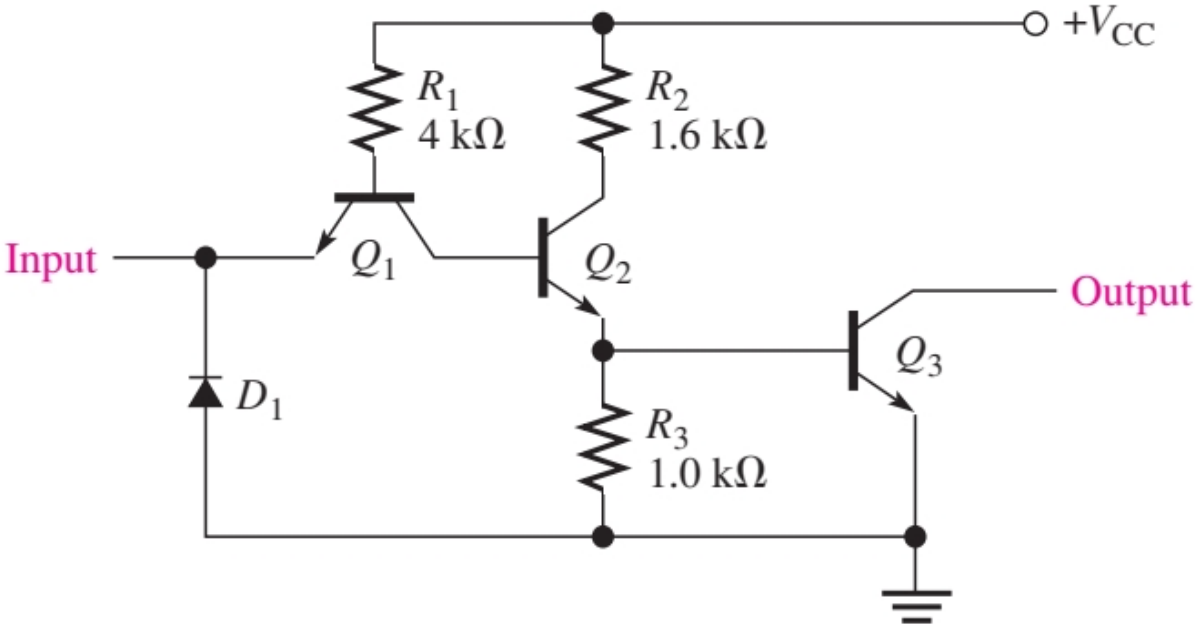
- TTL cổng NAND



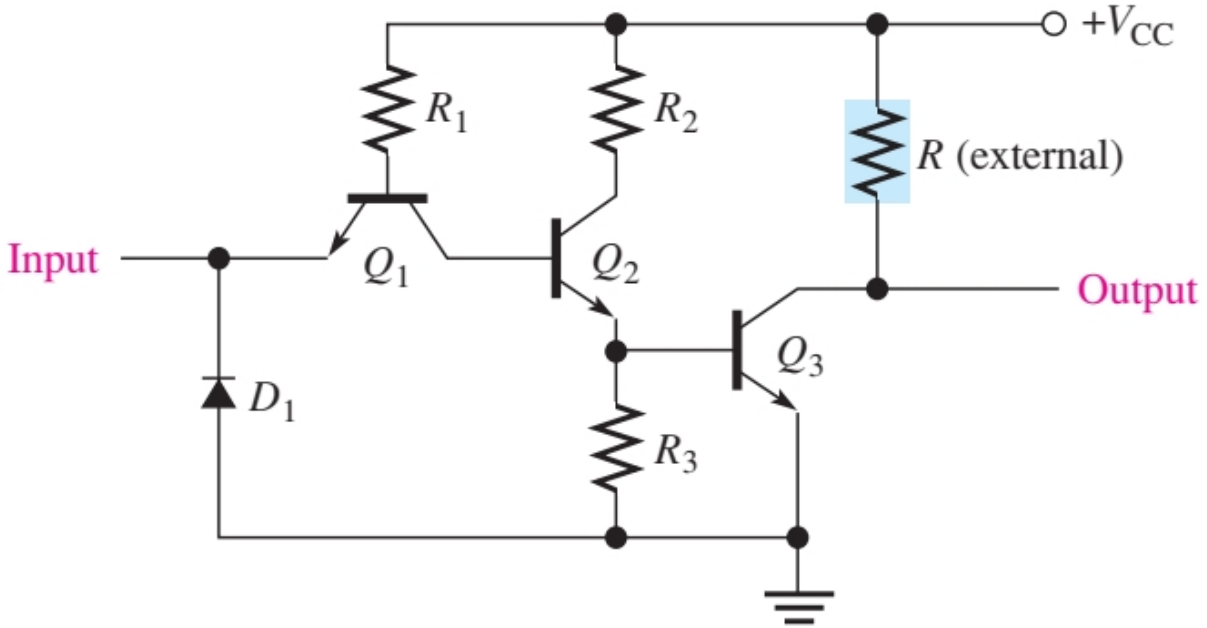


TTL

- TTL cực C hở



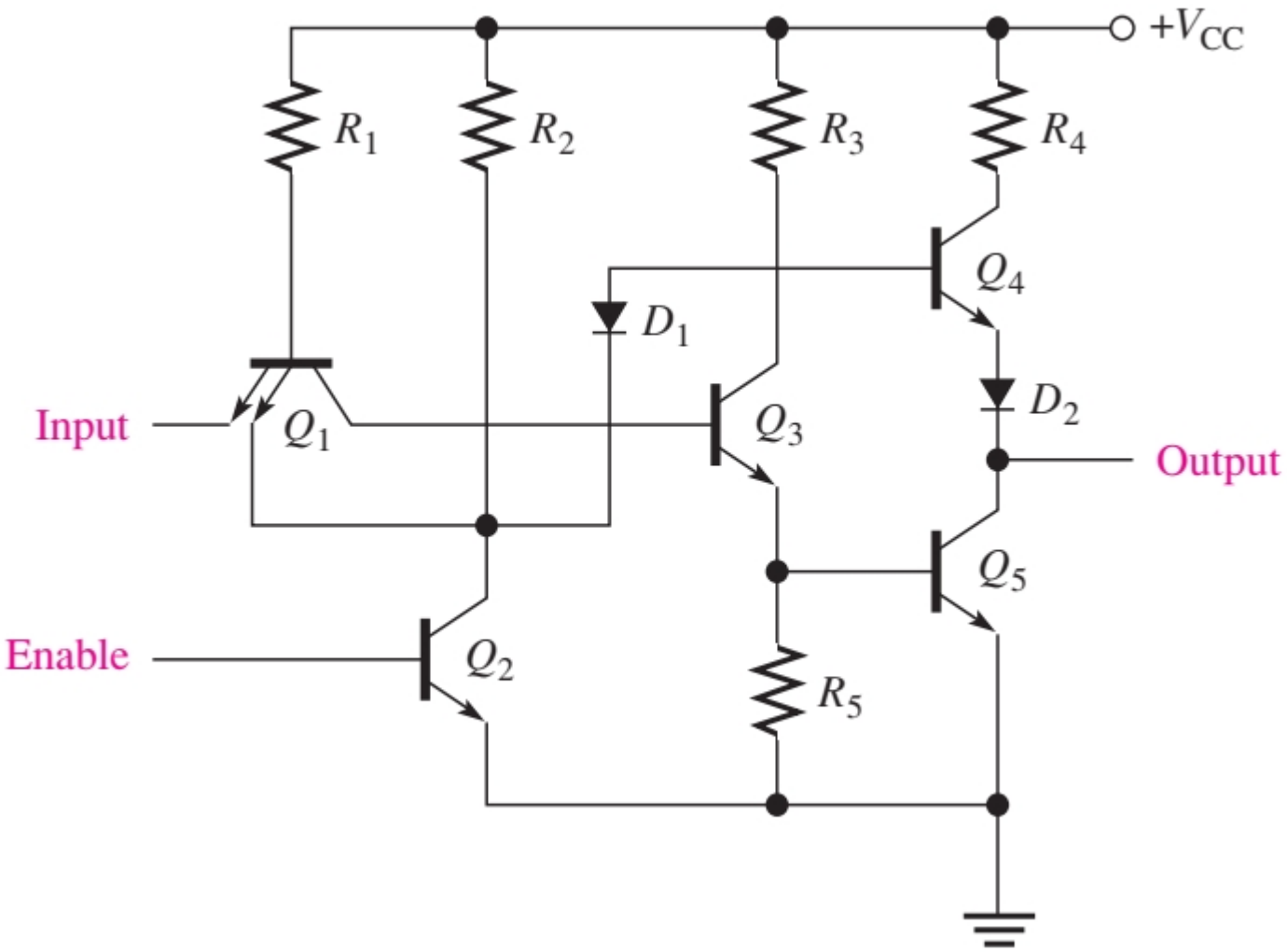
(a) Open-collector inverter circuit



(b) With external pull-up resistor

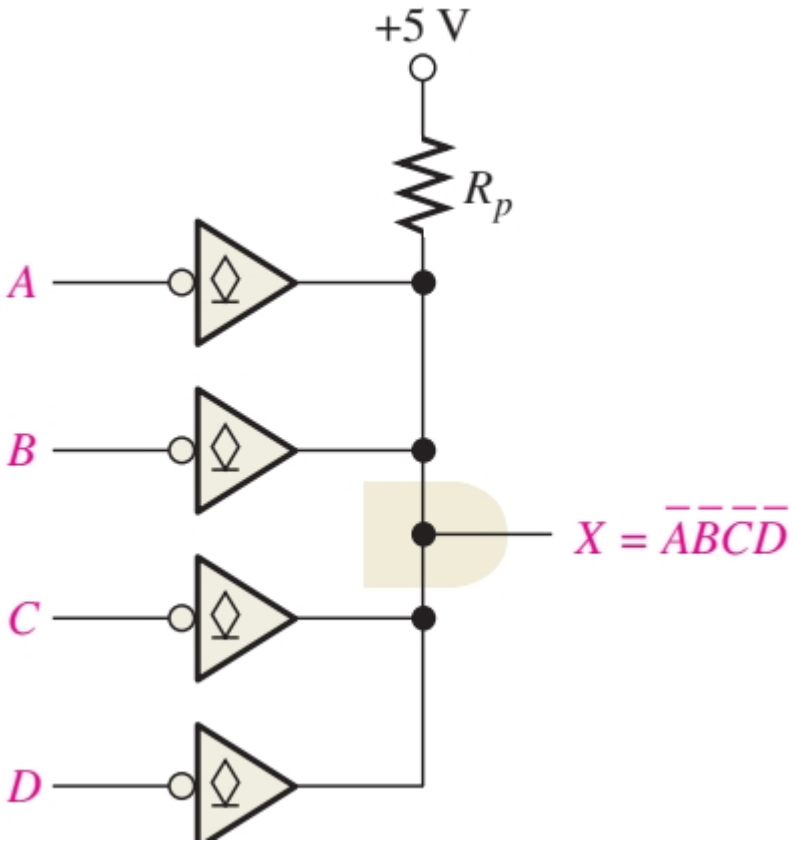
TTL

- TTL cổng ba trạng thái



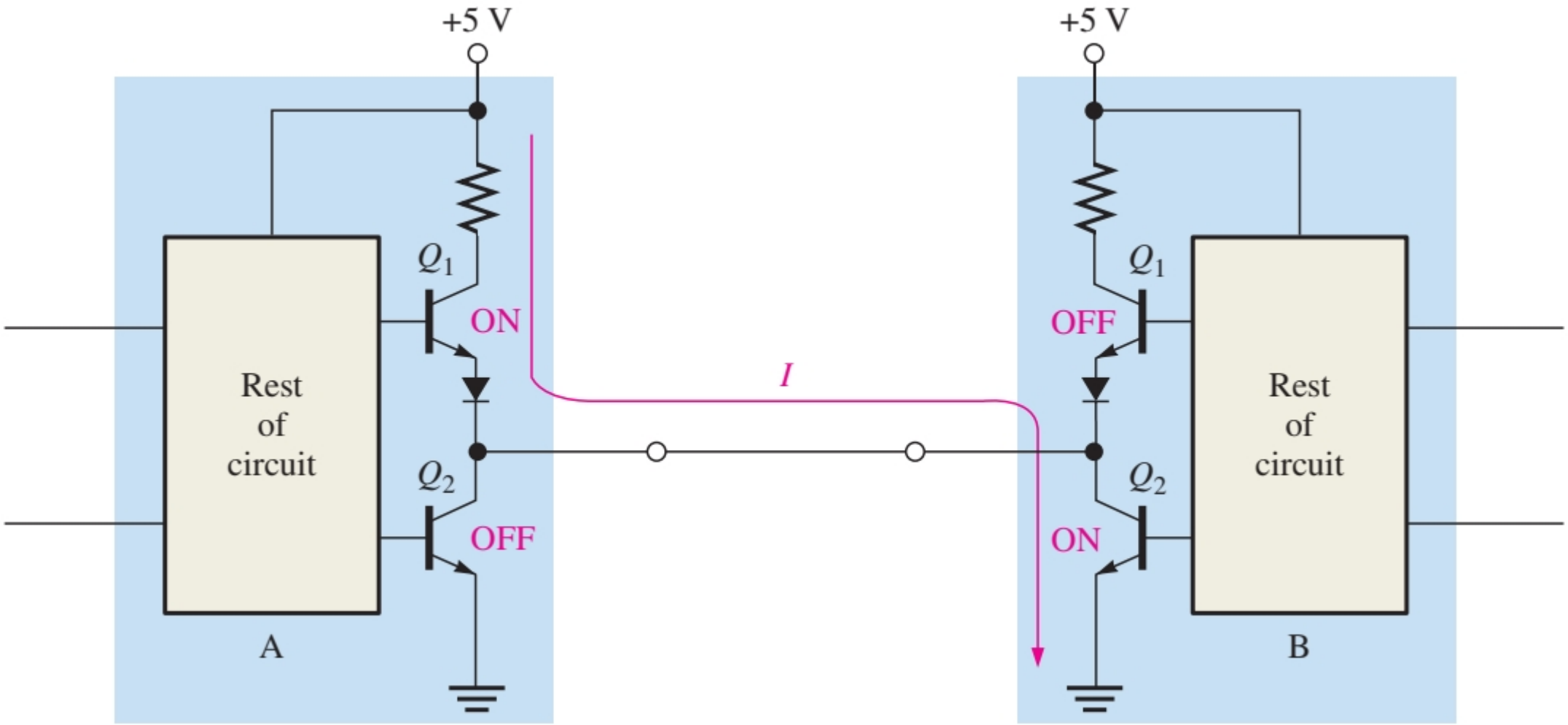
TTL

- TTL cổng wired-AND



TTL

- TTL kết nối Totem-Pole

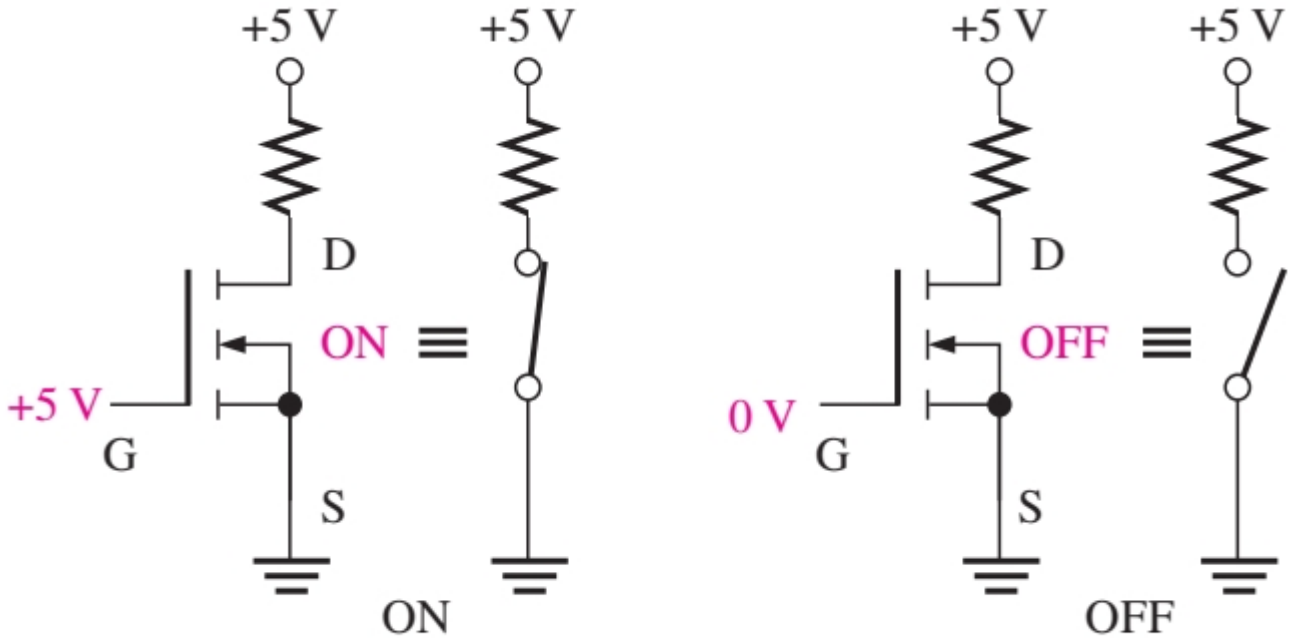


TTL

- TTL kết nối Totem-Pole

CMOS

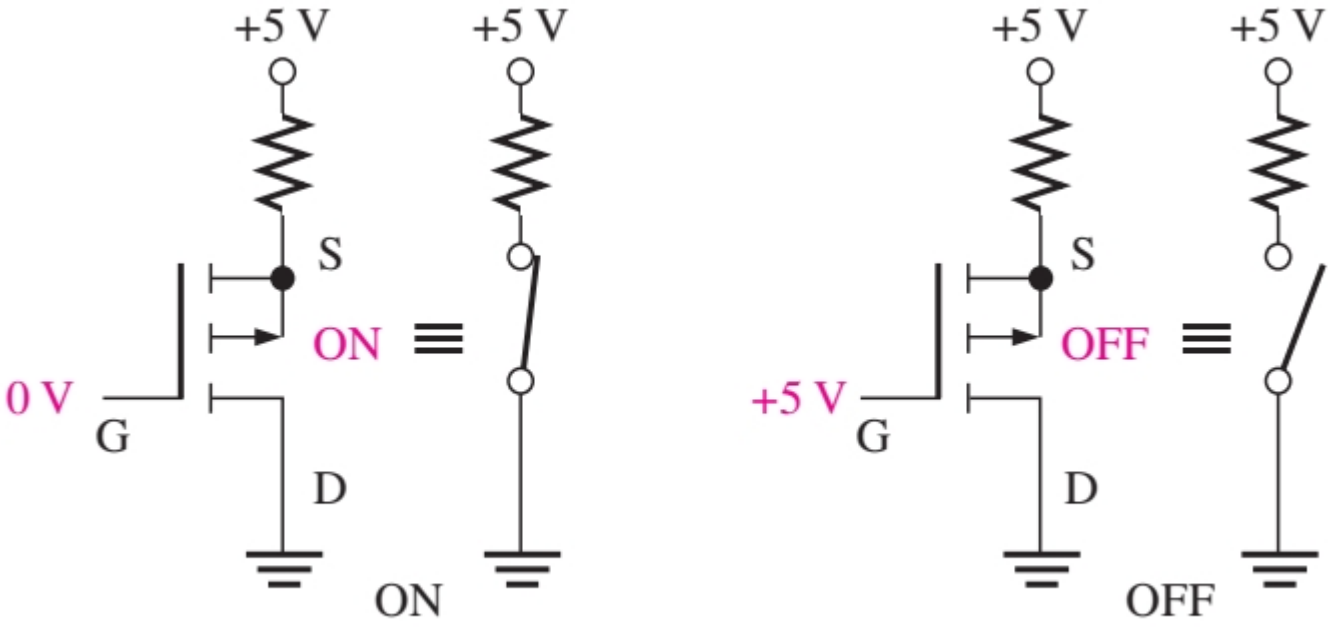
- MOSFEET



(b) *n*-channel switch

CMOS

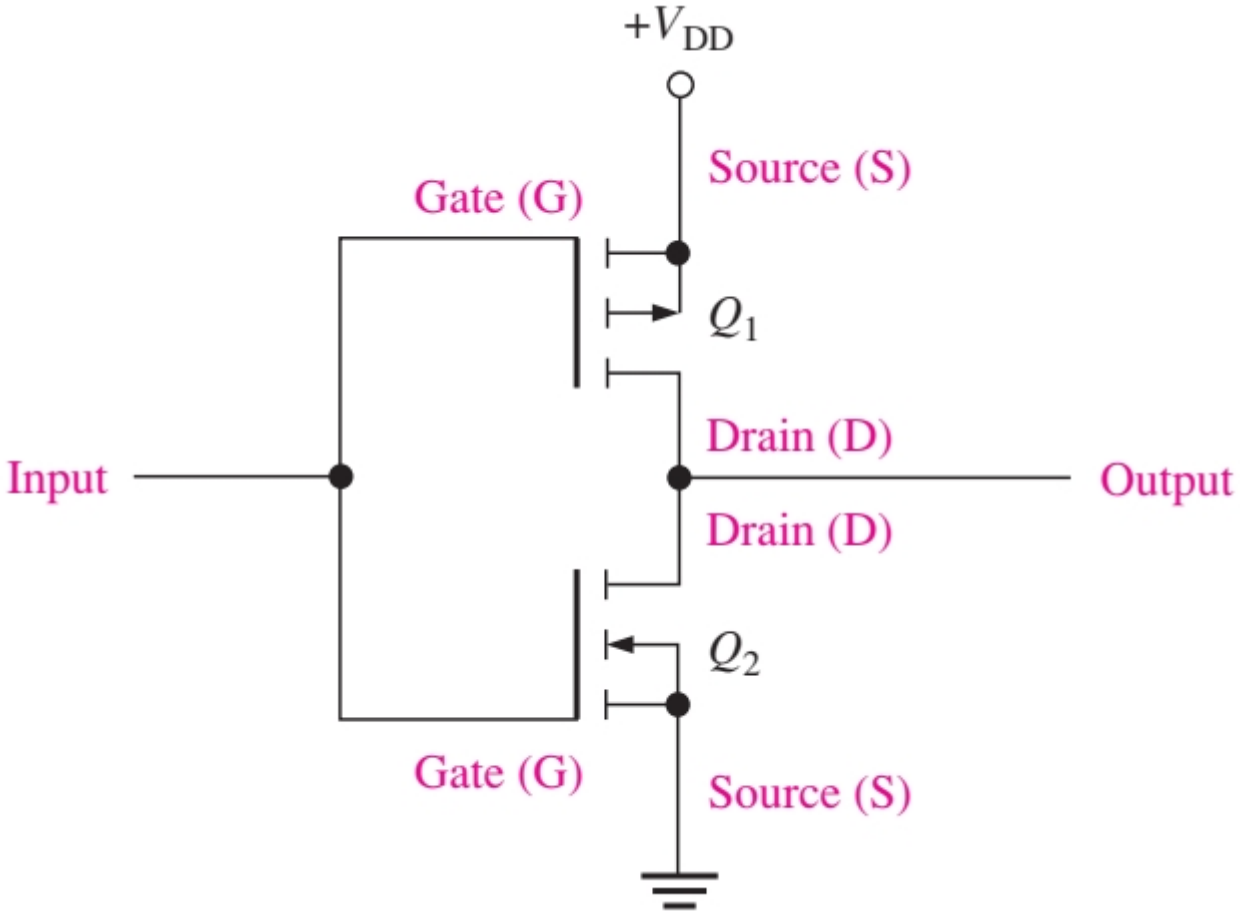
- MOSFEET



(c) *p*-channel switch

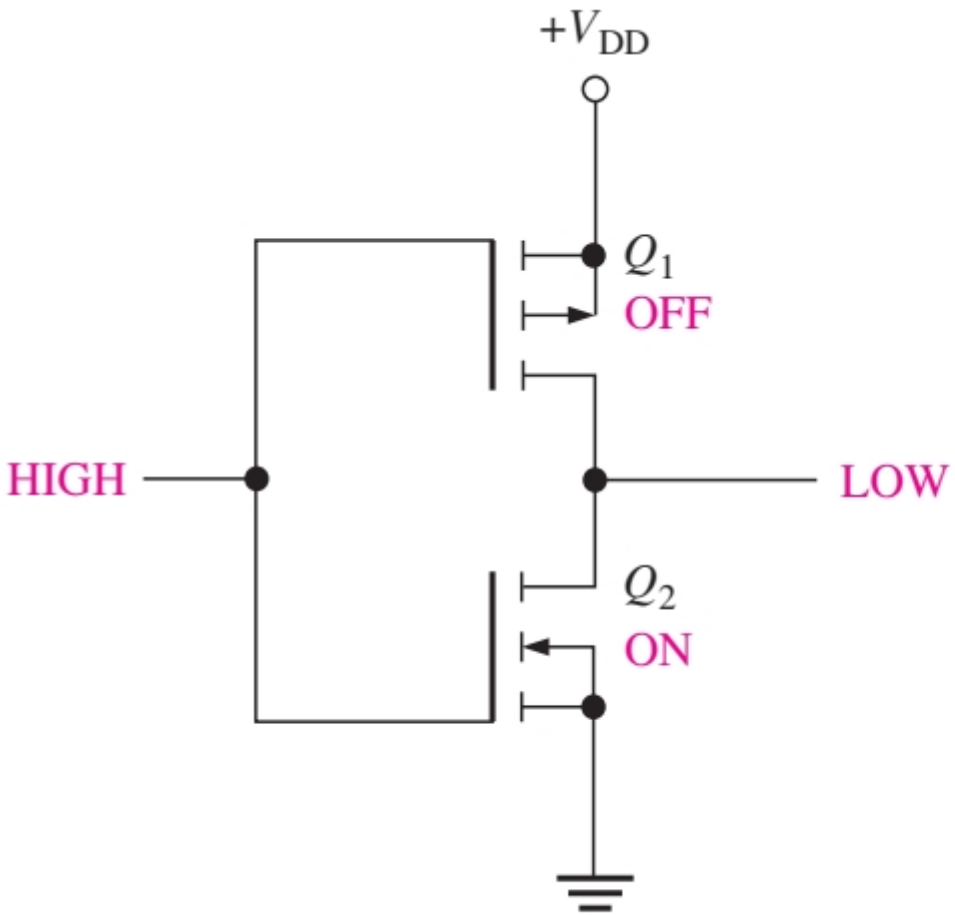
CMOS

- CMOS cổng NOT

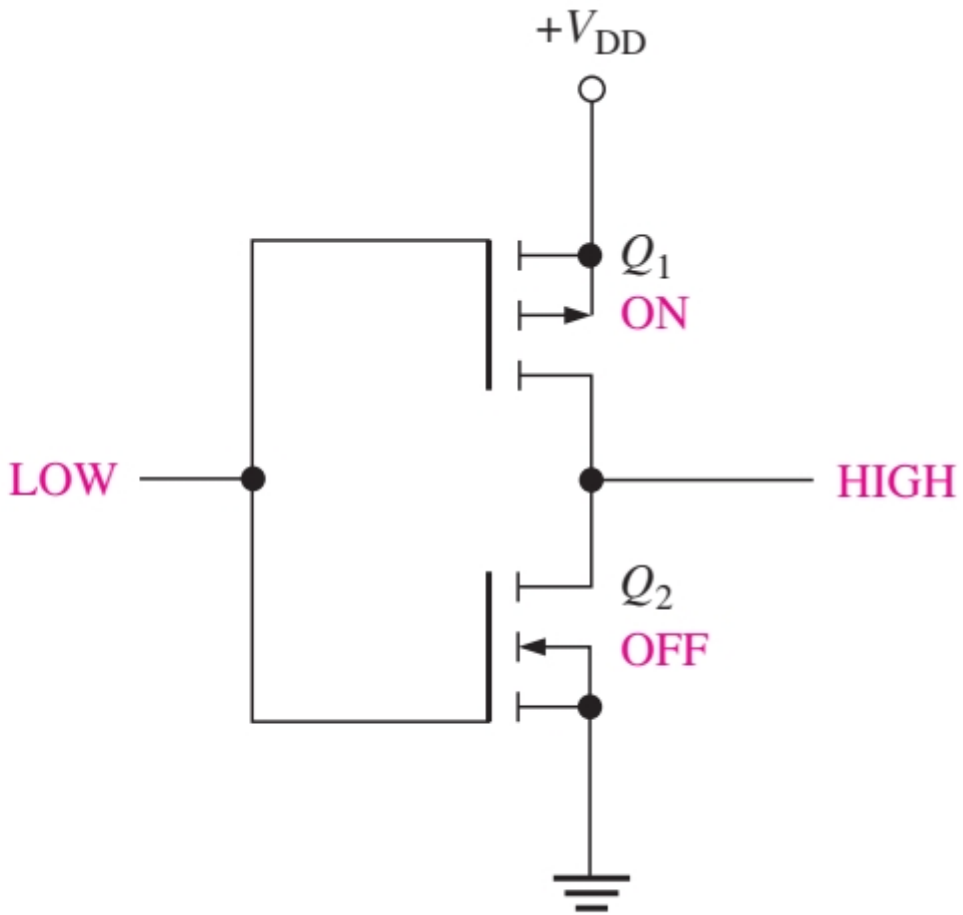


CMOS

- CMOS cổng NOT



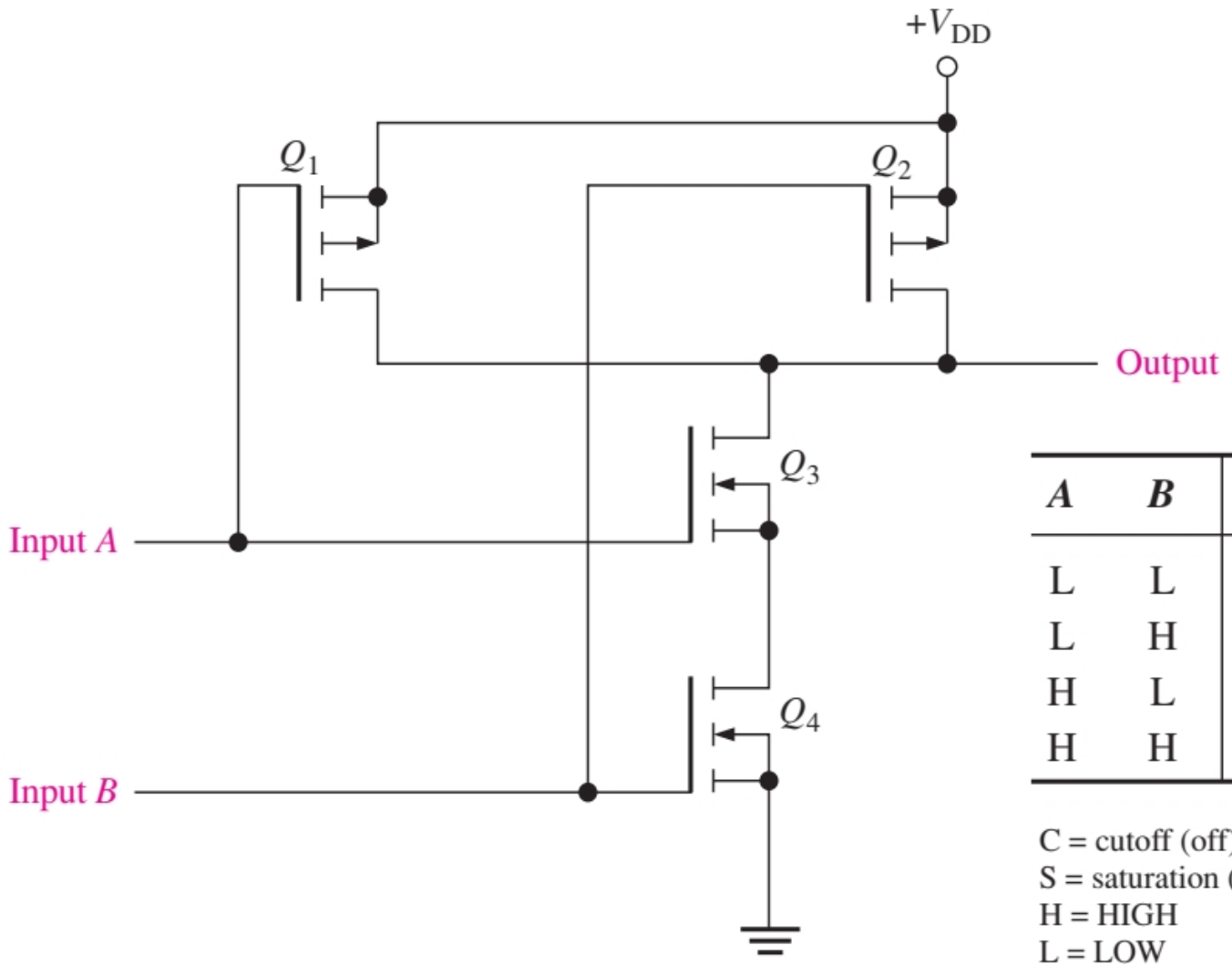
(a) HIGH input, LOW output



(b) LOW input, HIGH output

CMOS

- CMOS cổng NAND

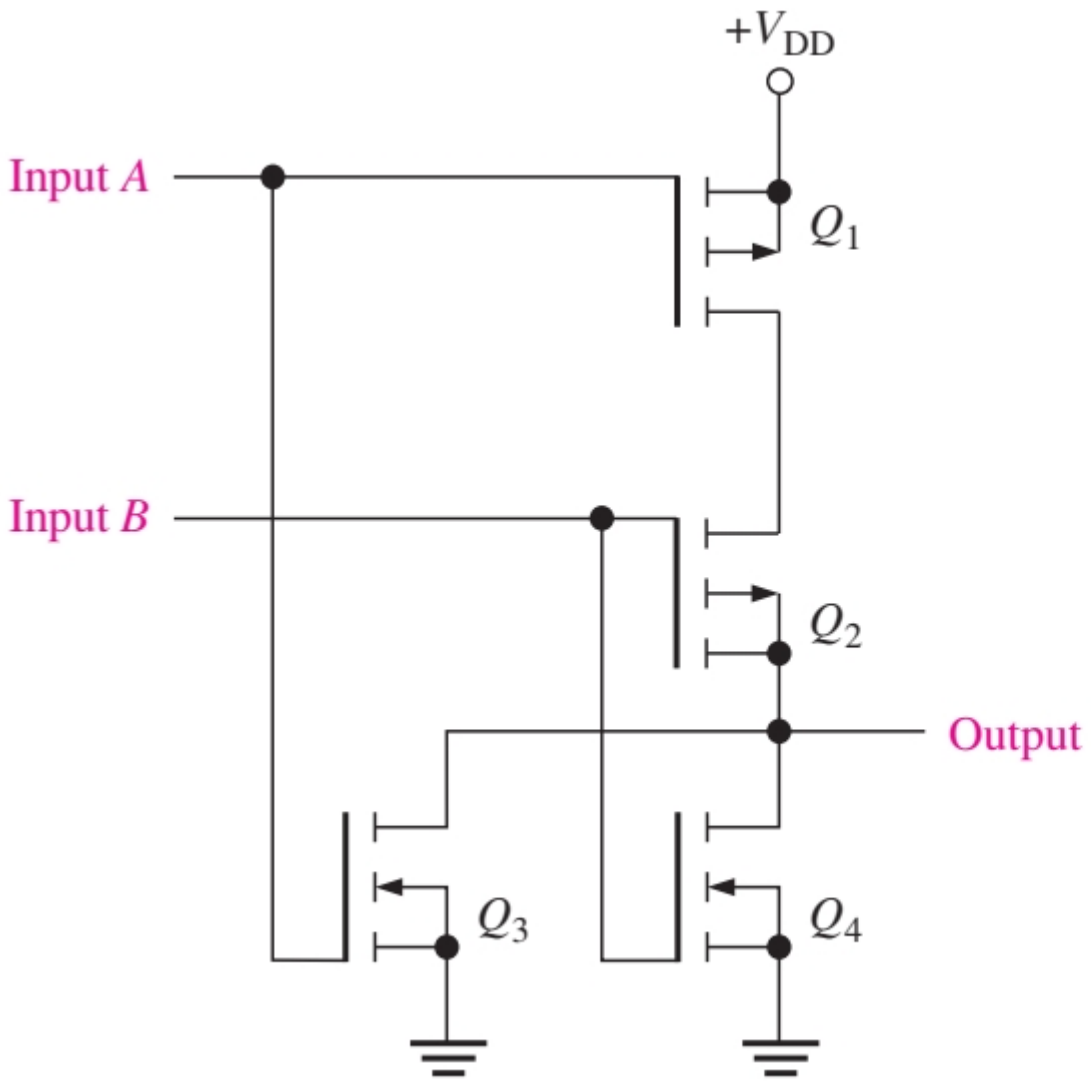


A	B	Q_1	Q_2	Q_3	Q_4	X
L	L	S	S	C	C	H
L	H	S	C	C	S	H
H	L	C	S	S	C	H
H	H	C	C	S	S	L

C = cutoff (off)
S = saturation (on)
H = HIGH
L = LOW

CMOS

- CMOS cổng NOR

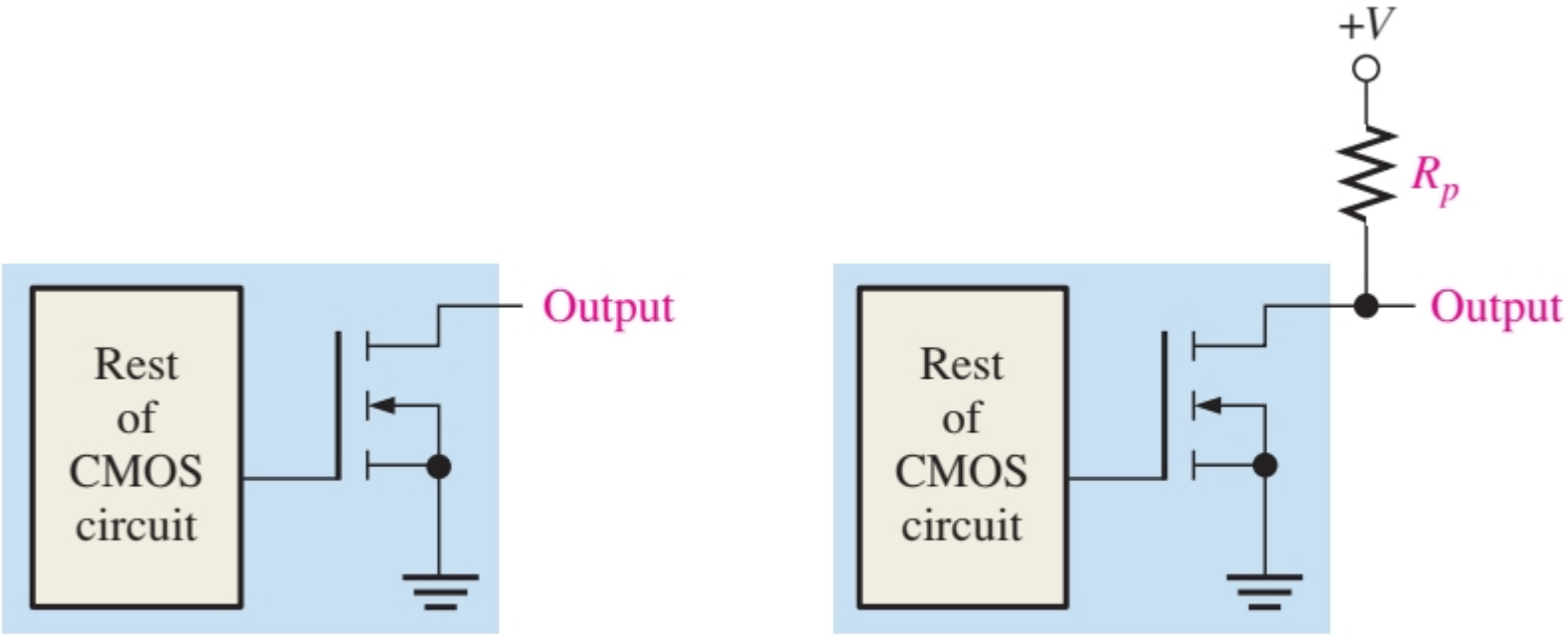


A	B	Q_1	Q_2	Q_3	Q_4	X
L	L	S	S	C	C	H
L	H	S	C	C	S	L
H	L	C	S	S	C	L
H	H	C	C	S	S	L

C = cutoff (off)
S = saturation (on)
H = HIGH
L = LOW

CMOS

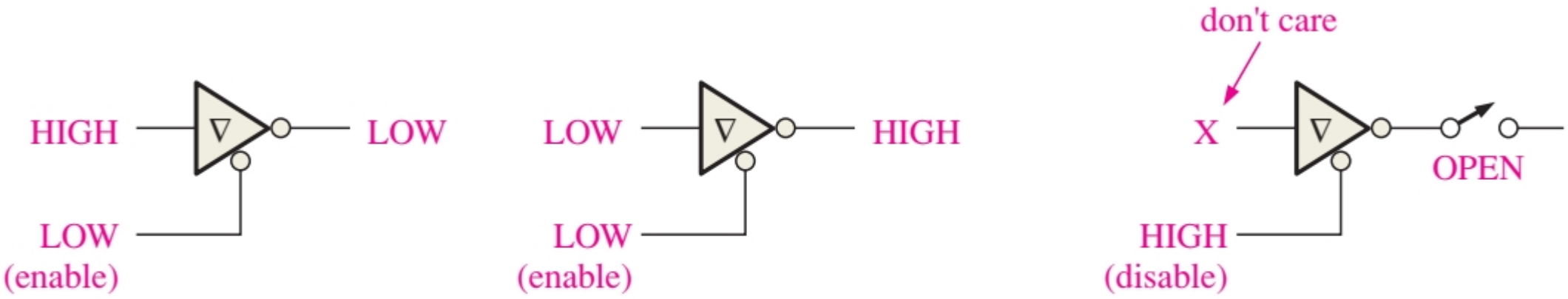
- CMOS cực D hở



(a) Unconnected output (b) With pull-up resistor

CMOS

- CMOS cổng ba trạng thái

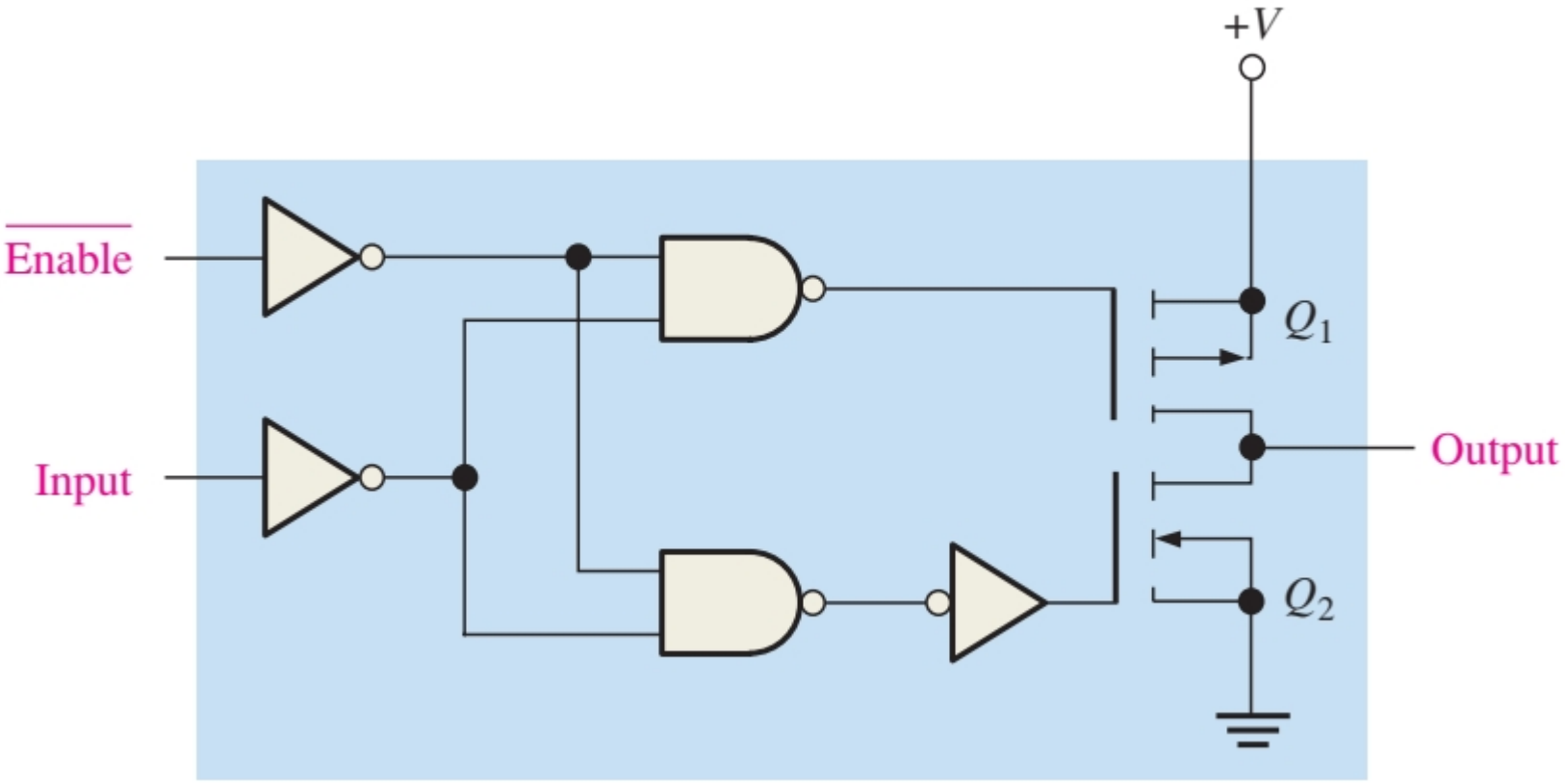


(a) Enabled for normal logic operation

(b) High-Z state

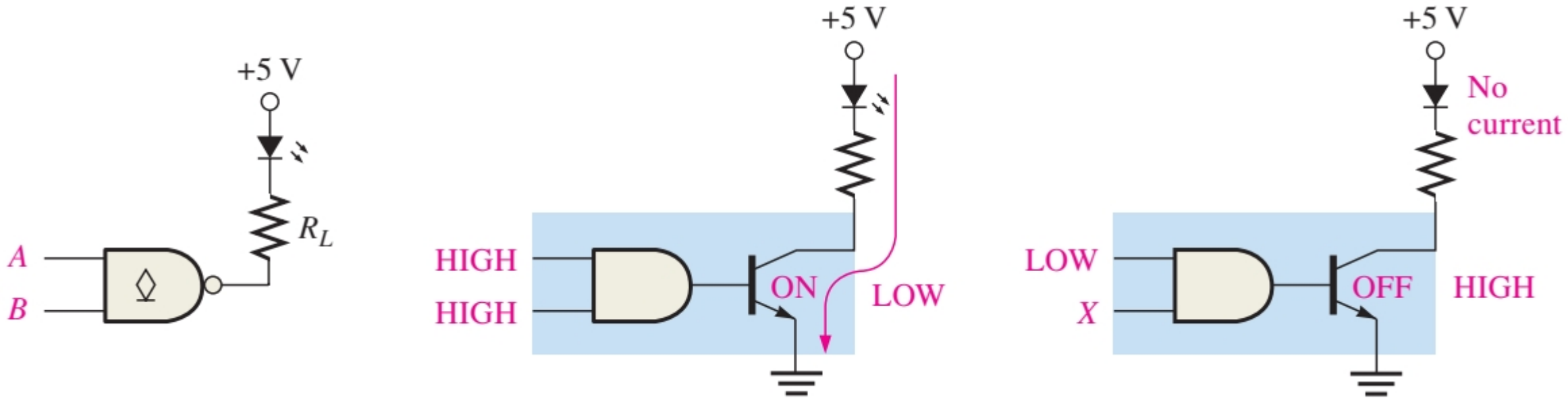
CMOS

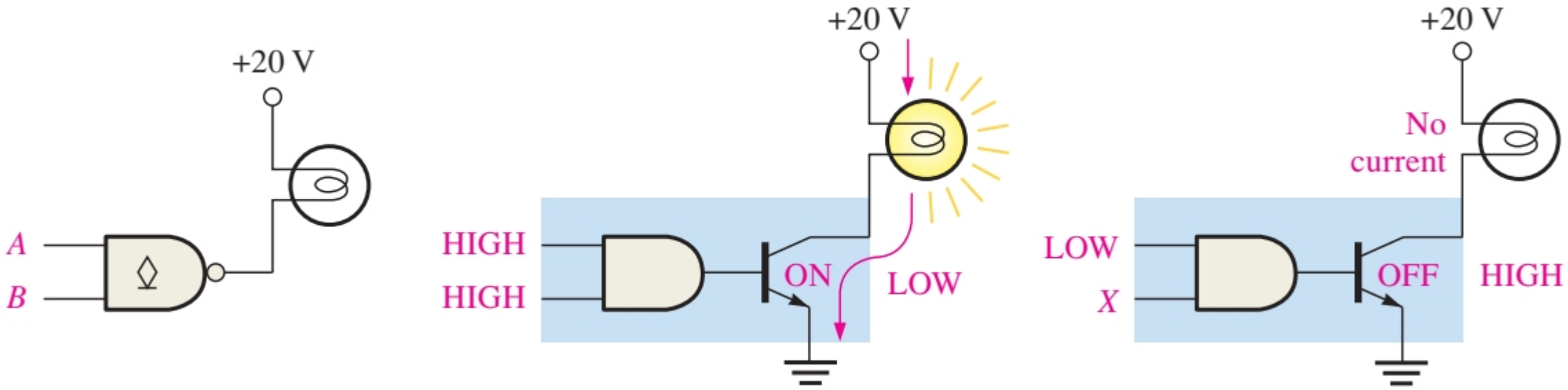
- CMOS cổng NOT ba trạng thái



Giao tiếp

- Cổng logic giao tiếp tải



Giao tiếp

- Cổng logic giao tiếp tải